00 · Start here: the whole lab on one page

ECE2130 / VDT2130 Electronic Devices Lab-I · source: the lab manual (session 2025-26), 39 pp. · every expected number in this pack comes from running the manual's own circuits in LTspice

The lab as one path

Instruments (Exp 0) → Diodes: 1 zener curve · 2 regulator · 3 clipper · 4 clamper →
BJT: 5 curves · 6 CE amp · 7 follower · 8 feedback → JFET: 9 curves · 10 CS amp → 11 oscillator →
12 MOSFET inverter → Extra: two-stage amp + design problem

Each block builds on the one before it:

- **Diodes first.** A diode is one junction. Exp 1 measures it, and Exps 2–4 use it in three circuits.
- **Then the BJT.** A transistor is two junctions. Exp 5 measures its curves, and Exps 6–8 turn it into amplifiers.
- **Then FETs.** The same "measure the curve, then build the amplifier" pattern, with a field-controlled device.
- **Exp 11 and Exp 12** take an amplifier somewhere new: positive feedback makes an oscillator, and a transistor used as a switch makes a logic gate.

How each experiment is run and marked

Exp	Topic	Done in	Marked by
0	DC and AC measurement	hands-on	not assessed
1	Zener characteristics, dynamic resistance	hands-on	viva, ETE
2	Zener line and load regulation	LTspice / Virtual Lab	viva, ETE
3	Clipper	hands-on	viva, ETE
4	Clamper	LTspice	viva, ETE
5	BJT CE characteristics, h-parameters	LTspice / Virtual Lab	viva, ETE
6	CE amplifier frequency response	hands-on	viva, ETE
7	Emitter follower gain	hands-on	viva, ETE
8	Negative feedback (bypass capacitor)	LTspice	viva, ETE
9	JFET drain and transfer characteristics	LTspice	viva, ETE
10	CS amplifier frequency response	LTspice	viva, ETE
11	RC phase-shift oscillator	LTspice	viva, ETE
12	MOSFET inverter	LTspice	viva, ETE

The manual does not give this lab's marks split. The Digital Electronics lab manual gives 100 = internal 60 + external 40, with a 3-hour exam. Expect the same shape here, but it is not confirmed.

ETE (end-term exam) for a lab usually means: you are given one experiment, you build or simulate it, you show the result, and you answer **viva** (oral) questions on it. So for every experiment you need to be able to:

1. draw the circuit from memory,
2. build or simulate it without help,
3. get the expected readings and calculate the result,
4. answer "why" questions about it.

Each chapter below gives you those four things, in that order.

What goes in the record (lab file)

The manual (p. 3) asks for six headings per experiment:

Heading	What to write
Aim	as in the list above
Circuit diagram	the circuit, labelled with values
Theory	about 15 lines. Each chapter has a ready "Theory for the record" block.
Observations	the table or a well-labelled waveform
Calculations	if any, with the formula first
Result	the parameters, **with % error**

$$\% \text{ error} = \frac{|\text{theoretical} - \text{observed}|}{\text{theoretical}} \times 100$$

How a chapter works

> **THE QUESTION**
> Each chapter opens with the experiment's aim, quoted from the manual, plus a real viva-style question.

> **GUESS FIRST**
> Answer it on paper before reading on, even if your answer is a wild guess. Guessing first makes the explanation stick better than reading cold. This line is the reason the pack works, not decoration.

Then come **the ideas needed**, **the circuit**, a **Do it** box with the exact steps, **what you should see** (simulated curves and numbers), the worked **calculation**, a **Write in your record** box, **traps**, and **checks**. The answers to the checks are on the last page of each chapter.

Errors in the manual (know these for the viva)

Where	The manual says	Correct
Exp 0	$V_{rms} = V_p/\sqrt{2}$, with V_p read as peak-to-peak	For a sine, $V_{rms} = V_{peak}/\sqrt{2} = V_{pp}/(2\sqrt{2})$
Exp 1	Requirement: diode 1N4007	The experiment needs a **zener** diode. A 1N4007 is an ordinary rectifier and will not show a sharp breakdown below about 1000 V.
Exp 5	h_{21} = output admittance, h_{22} = forward current gain	Swapped. $h_{21} = h_{fe}$ (forward current gain), $h_{22} = h_{oe}$ (output admittance).
Exp 5	$1/h_{oe} = \Delta I_C/\Delta V_{CE}$	$h_{oe} = \Delta I_C/\Delta V_{CE}$. The ratio is a conductance, so $1/h_{oe}$ is the output resistance.
Exp 5	"Keep emitter current I_b = 20 A"	**Base** current I_B = 20 **µA**
Exp 5	Fig 5.1 with 62 kΩ in the collector	With 62 kΩ, I_C can never exceed 10 V / 62 kΩ = 0.16 mA, so the I_B = 20/40 µA table cannot be filled. See Exp 5.
Exp 6	Input 500 mV	This clips the output badly. Use about 20 mV (Exp 6 shows both).
Exp 7	Input impedance "of the order of hundred ohms"	Input impedance is **high**: about 3 kΩ here (set by the bias resistors), and about 600 kΩ looking into the base alone. The **output** impedance is the low one (about 70 Ω).
Exp 9	$r_d = (7 - 6)/(0.1183 - 0.1180) = 3.3$ kΩ	Decimal slip. The manual's own data (11.83 − 11.80 mA) gives 1 V / 0.03 mA = **33 kΩ**. LTspice gives 38 kΩ.
Exp 12	"V_{th} is the inverter threshold voltage, which is $V_{dd}/2$"	Two different things. V_{TH} is the MOSFET's turn-on voltage. V_M is where $V_{out} = V_{in}$. $V_M = V_{DD}/2$ only for a symmetric CMOS inverter, not for this resistor-load circuit ($V_M \approx 0.76$ V here).

01 · LTspice from zero

Needed for Exps 2, 4, 5, 8, 9, 10, 11, 12 and the extra experiment · LTspice is free (Analog Devices) and already installed on your laptop

Where this sits

LTspice basics → the 4 analyses → drawing a circuit → reading the result → exporting data

What you already have: circuits from C&S (KVL, KCL, sources, R, C), and the idea that a graph of one quantity against another describes a device.

Three terms first:

- **Simulator:** a program that solves the circuit's KVL/KCL equations for you, using a mathematical **model** of each part.
- **Model:** the set of numbers that describes a real part. For example, "BC547B" tells LTspice the β, the junction capacitances, and so on.
- **Directive:** a text command starting with a dot (`.dc`, `.ac`, `.tran`, `.op`, `.step`). It tells LTspice *which* analysis to run.

MANUAL EXP 2, FIG 2.1

The circuit has `.dc V1 0 20 1` written under it. What does that line make LTspice do?

GUESS FIRST

Guess what each of the four words means: `.dc`, `V1`, `0 20`, `1`.

The four analyses (every software experiment uses one or more)

Directive	What it does	Gives you	Used in
`.op`	finds the DC operating point (all capacitors open)	one number per node	Exp 2 (load), checking a Q-point
`.dc V1 0 20 1`	sweeps source V1 from 0 to 20 V in 1 V steps, solving at each step	a curve: anything against V1	Exps 2, 5, 9
`.ac dec 10 200 1Meg`	small-signal sine at each frequency, 10 points per decade, 200 Hz to 1 MHz	gain (dB) and phase against frequency	Exps 7, 8, 10, extra
`.tran 10m`	runs time forward for 10 ms	waveforms against time	Exps 4, 7, 10, 11, 12
`.step param R 100 1100 100`	reruns the analysis for R = 100, 200, … 1100	a family of curves	Exp 2 (load)

So `.dc V1 0 20 1` = "DC sweep of V1, start 0, stop 20, step 1". A **nested** sweep has a second source at the end: `.dc Vce 0 12 0.05 Ib 0 60u 20u` sweeps V_{ce} for each of I_b = 0, 20, 40, 60 µA, which gives one curve per I_b.

Units: the trap that costs marks

SPICE ignores case, so `M` **means milli, not mega.**

You want	Type	Not
1 MΩ	`1Meg`	~~`1M`~~ (that is 1 mΩ)
10 µF	`10u`	
0.5 nF	`0.5n` or `500p`	
2.2 kΩ	`2.2k`	

Drawing a circuit

DO IT: DRAW, SET, RUN

1. **File → New Schematic.**
2. Place parts using the keyboard: **R** resistor, **C** capacitor, **D** diode, **G** ground. **F2** opens the full component list (`voltage`, `npn`, `njf`, `nmos`, `zener`).
3. **F3** draws a wire: click at the start, click at each corner, right-click to stop. **Ctrl+R** rotates a part before you place it.
4. **Every circuit needs a ground (G).** Without one, the run fails with a "singular matrix" or "floating node" error.
5. **Right-click a part** to set its value (e.g. `2.2k`).
6. To choose a **model** (BC547B, 2N3819, 1N914, BZX84C6V2L, FDR6580, 2N2222, 2N5434), right-click the diode or transistor and click **Pick New ...**. All the manual's parts are in LTspice's library.
7. To set up a **source**, right-click it and click **Advanced**: - **SINE**: DC offset, amplitude, frequency. `SINE(0 5 1k)` = 0 V offset, 5 V peak, 1 kHz. - **Small signal AC amplitude**: type `1` (or `0.5`) here for an `.ac` run. - **PULSE**: V1, V2, delay, rise, fall, on-time, period, cycles.
8. To add a directive, press **S** (or **Edit → SPICE Directive**), type it, and place the text anywhere on the sheet. Alternatively, **Simulate → Edit Simulation Cmd** fills the directive in for you.
9. **F4** puts a label on a wire. Name the output `Vo` so it is easy to plot.
10. Click **Run** (the running-man icon).

Reading the result

You want	Do this
a node voltage	after the run, click on the wire (the probe turns into a red voltage probe)
a current	click on the part's body (the probe turns into a current clamp)
a different x-axis (e.g. V(base) instead of V_{be})	right-click the x-axis, then set **Quantity plotted** to `V(base)`
an exact value	click the trace's name above the plot. A cursor appears, and its readout box shows x and y. A second click adds cursor 2; the box then also shows the difference.
gain in plain ratio, not dB	right-click the y-axis and choose **Linear**
y-axis limits	right-click the y-axis and set **Manual limits**

In an `.ac` plot, the **solid line is the gain in dB** and the **dashed line is the phase**.

OBSERVATION TABLES FROM A SIMULATION

The manual says: "Paste data points from simulations". With the plot window active, use **File → Export data as text**. The file has one row per point. Copy only the rows the table asks for (5–10 points), and mark on the graph the points used in calculations.

TRAP: THE DEFAULT TIME STEP

`.tran 20m` lets LTspice pick its own step, which can be coarse. Edges then look slanted, or tall false spikes appear. Write `.tran 0 20m 0 1u` instead: the fourth number caps the step at 1 µs.

CHECK 1

(a) Write the directive that sweeps V_{gs} from −5 V to 0 V in 10 mV steps. (b) Your 1 MΩ resistor shows `1M`. What value is LTspice actually using? (c) Which analysis gives a frequency response?

Answers

Check 1. (a) `.dc Vgs -5 0 0.01` (or `10m`). (b) 1 milliohm, a short circuit. Write `1Meg`. (c) `.ac`, with the source's AC amplitude set.

Opening question. `.dc` = DC sweep, `V1` = the source being swept, `0 20` = from 0 V to 20 V, `1` = in 1 V steps. LTspice solves the circuit 21 times and plots any node against V1.

Exp 0 · Measuring DC and AC: power supply, function generator, DSO, multimeter

Manual pp. 4–6 · hands-on · not assessed on its own, but every hands-on experiment (1, 3, 6, 7) depends on it

0 Instruments → 1 Zener curve → 2 Regulator → 3 Clipper → 4 Clamper

What you already have: voltage, current, and a sine wave $v(t) = V_p \sin(2\pi ft)$ from C&S.

Three terms first:

- **Peak V_p:** the sine's height above its centre line.
- **Peak-to-peak V_{pp}:** top to bottom, $= 2V_p$.
- **RMS:** the steady DC value that would heat a resistor equally. For a sine, $V_{rms} = V_p/\sqrt{2}$.

MANUAL EXP 0, AIM

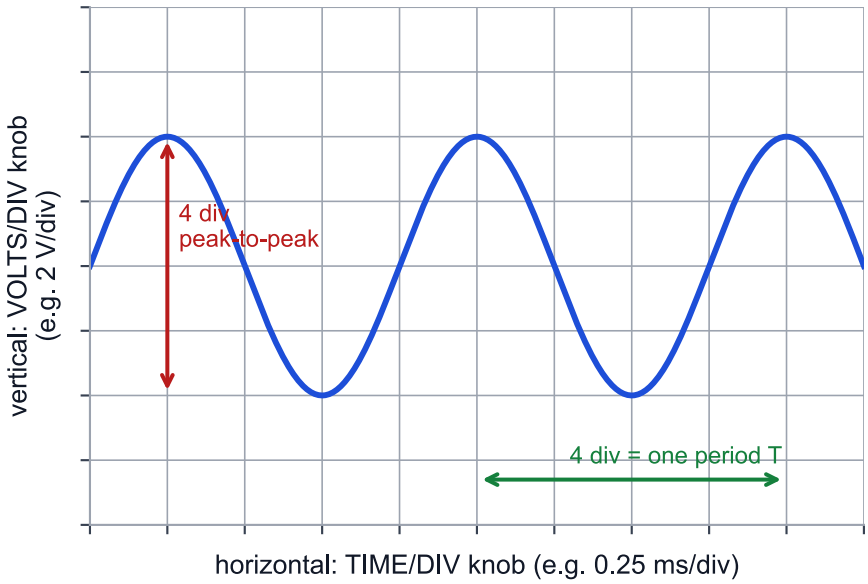
Generation and measurement of the DC voltage and AC signals. Viva form: "The DSO shows 4 divisions peak-to-peak at 2 V/div, and one cycle spans 4 divisions at 0.25 ms/div. Give V_{pp}, V_{rms} and f."

GUESS FIRST

Write your three numbers before reading on.

The four instruments

Instrument	What it does	The knobs that matter
DC power supply	gives a steady voltage	voltage knob; **current-limit** knob (set it low to protect the circuit); + and − terminals
Function generator	gives a sine, square or triangle wave	waveform buttons, frequency range + fine knob, **amplitude**, **DC offset** (keep at 0 unless asked)
DSO (digital storage oscilloscope)	draws voltage against time	**VOLTS/DIV** (vertical scale), **TIME/DIV** (horizontal scale), **coupling DC/AC**, trigger, **Auto-set**, **Measure** menu
DMM (digital multimeter)	gives one number: V, A or Ω	knob position; black lead in **COM**; red lead in **VΩmA** (or **10A** for currents above 200 mA)



Reading a DSO screen: count divisions, then multiply by the knob setting.

Reading the screen (the answer to the question)

- **Volts:** V_{pp} = (divisions top to bottom) × (VOLTS/DIV) = 4 × 2 = **8 V**. So V_p = 4 V.
- **RMS of a sine:** $V_{rms} = V_p/\sqrt{2} = 4/1.414 =$ **2.83 V**. This is what a DMM on AC volts should show.
- **Time:** T = (divisions per cycle) × (TIME/DIV) = 4 × 0.25 ms = 1 ms, so **f = 1/T = 1 kHz**.

Wave	V_{rms}
sine	$V_p/\sqrt{2} = 0.707\ V_p$
triangle	$V_p/\sqrt{3} = 0.577\ V_p$
square	V_p

TRAP: THE MANUAL'S RMS FORMULA

The manual writes $V_{rms} = V_p/\sqrt{2}$ but has you read "V_p" as **peak-to-peak**. Divide **peak** by √2, or peak-to-peak by $2\sqrt{2}$. Using $V_{pp}/\sqrt{2}$ gives an RMS twice too large.

TRAP: THE METER AND THE WAVE SHAPE

A basic DMM is calibrated for **sine waves**. On a square or triangle wave it can read wrong unless it says "True RMS". Basic meters also lose accuracy at higher frequencies, so keep the generator at about 1 kHz or lower for the meter comparison.

DO IT (HANDS-ON)

DC:

1. Set the supply to a value (e.g. 5 V), with the current limit low.
2. Connect the DSO probe tip to + and the probe's ground clip to −. Coupling **DC**.
3. Count the divisions the trace rises from the 0 V line, and multiply by VOLTS/DIV.
4. Read the same output with the DMM on DC volts.

AC:

1. Connect the generator output to the DSO (probe tip to signal, ground clip to ground).
2. Press Auto-set, or set VOLTS/DIV and TIME/DIV so that 1–2 cycles fill the screen.
3. Count V_{pp} divisions and the divisions in one period T. Calculate V_{rms} and f.
4. Measure the same output with the DMM on AC volts.
5. Repeat for the square and triangle waves.

WRITE IN YOUR RECORD

Observation table columns: wave shape · divisions for V_{pp} · VOLTS/DIV · V_{pp} · divisions for T · TIME/DIV · T · f = 1/T · V_{rms} (calculated) · V_{rms} (DMM).

Result: average error of the DSO reading = (|DMM − DSO| / DMM) × 100 averaged over the readings.

Theory for the record: a function generator produces sine, square and triangle voltages of set amplitude and frequency. A DSO samples the input, stores it digitally and draws voltage against time on a grid. Amplitude = divisions × VOLTS/DIV, and period = divisions × TIME/DIV, with f = 1/T. A DMM measures DC/AC voltage, current and resistance. On AC it displays RMS, which for a sine equals $V_{peak}/\sqrt{2}$. A regulated DC supply holds its output constant against changes in load and line.

CHECK 1

(a) A triangle wave is 6 V peak-to-peak. What is its RMS? (b) The probe's ground clip is left unconnected. What does the trace typically look like? (c) Why must the DMM be on the mA port, and in series, to measure a current?

Answers

Opening question. V_{pp} = 8 V, V_{rms} = 2.83 V, f = 1 kHz.

Check 1. (a) V_p = 3 V, so V_{rms} = 3/√3 = **1.73 V**. (b) A noisy, fuzzy trace (often showing 50 Hz mains pickup), because the probe has no reference point. (c) An ammeter must carry the same current as the branch, so it goes in series. It has a very low resistance, so it must never be placed across a source: that is a short circuit, and it blows the meter's fuse.

Exp 1 · Zener diode characteristics and dynamic resistance

Manual pp. 7–8 · hands-on · CO1, CO2 · theory: ECE2101 unit 2 (pn junctions)

0 Instruments → **1 Zener curve** → 2 Regulator → 3 Clipper → 4 Clamper

What you already have: Ohm's law, and the meters from Exp 0.

Three terms first:

- **Diode:** a p-type and an n-type piece of silicon joined together (a *pn junction*). The p side is the **anode** and the n side is the **cathode** (the bar on the symbol). Current flows easily from anode to cathode.
- **Forward / reverse bias:** anode more positive than cathode = forward. The reverse = reverse.
- **Breakdown:** under enough reverse voltage, a junction suddenly conducts heavily.

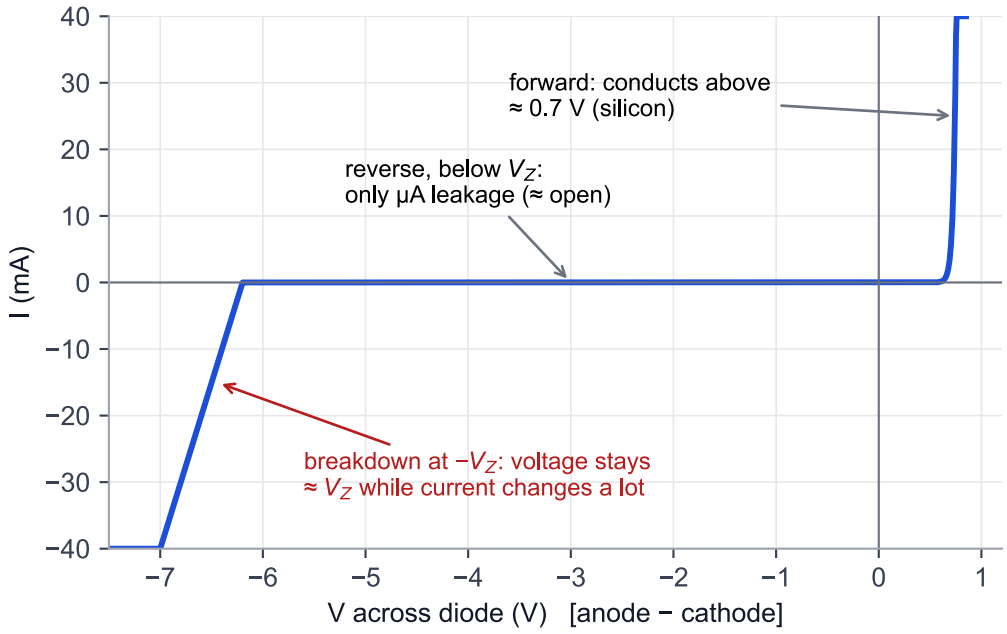
MANUAL EXP 1, AIM

To plot the characteristics of a zener diode and find its dynamic resistance under reverse-biased condition. Viva form: "Why does a zener's reverse current rise sharply at one voltage, and what is its *dynamic* resistance?"

GUESS FIRST

Sketch I against V for an ordinary diode, both directions, before reading on. Where does it conduct?

The idea: a diode that is meant to break down



One curve, three regions. A zener is built so the left-hand breakdown happens at a chosen, sharp voltage V_Z.

1. **Forward:** almost no current until about 0.6–0.7 V (silicon), then current rises steeply. This is the same as any diode.
2. **Reverse, below V_Z:** only a tiny leakage current (μA). The diode is close to an open circuit.
3. **Reverse, at V_Z:** the junction **breaks down**. The current can rise from 1 mA to 10 mA while the voltage barely moves from V_Z.

Region 3 is what makes a zener useful: **a voltage that stays put while the current changes**, which is a voltage reference. Exp 2 uses exactly that.

Why breakdown happens (viva staple; from ECE2101 unit 2):

	Zener breakdown	Avalanche breakdown
doping	heavy, so the junction is thin	light, so the junction is wide
mechanism	the field pulls electrons straight across (tunnelling)	fast carriers knock out new ones (chain reaction)
V_Z	below about 5 V	above about 6 V
temperature coefficient	negative	positive

A 6.2 V zener is near the crossover and uses both mechanisms. That is why ~5–6 V zeners drift least with temperature. "Zener diode" is the trade name for any breakdown diode, whichever mechanism it uses.

Dynamic resistance r_z: in breakdown, how much the voltage moves per unit change of current:

$$r_z = \frac{\Delta V_Z}{\Delta I_Z}$$

A small r_z (tens of ohms or less) means a better voltage reference. It is called *dynamic* because it is the slope of the curve, not V/I at one point. V_Z/I_Z would be a static resistance, and a meaningless one here.

Limits:

- **$I_{Z(min)}$:** the least current that keeps the zener in breakdown.
- **$I_{Z(max)} = P_Z/V_Z$:** the largest current before it overheats. A 0.5 W, 6.2 V zener allows $I_{Z(max)} = 0.5/6.2 = 81$ mA.

The series 1 kΩ resistor in the circuit limits the current.

The circuit

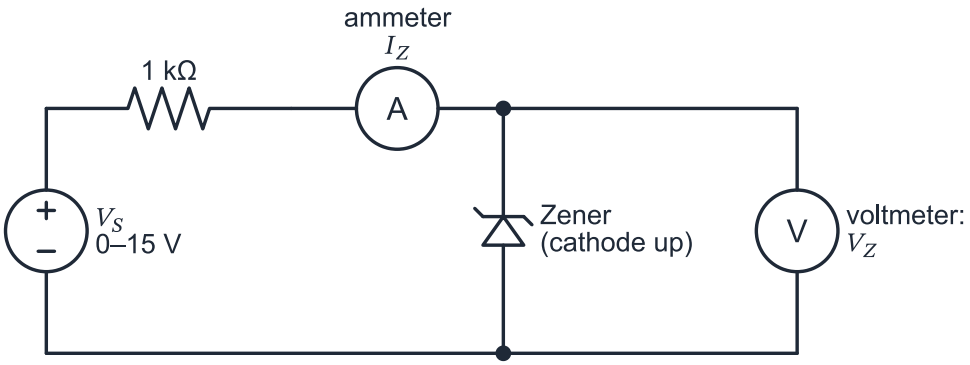


Fig 1.1, redrawn. The zener's cathode faces the + side, so it is reverse-biased. The ammeter is in series; the voltmeter (or DSO) is across the zener.

TRAP: THE MANUAL'S PARTS LIST

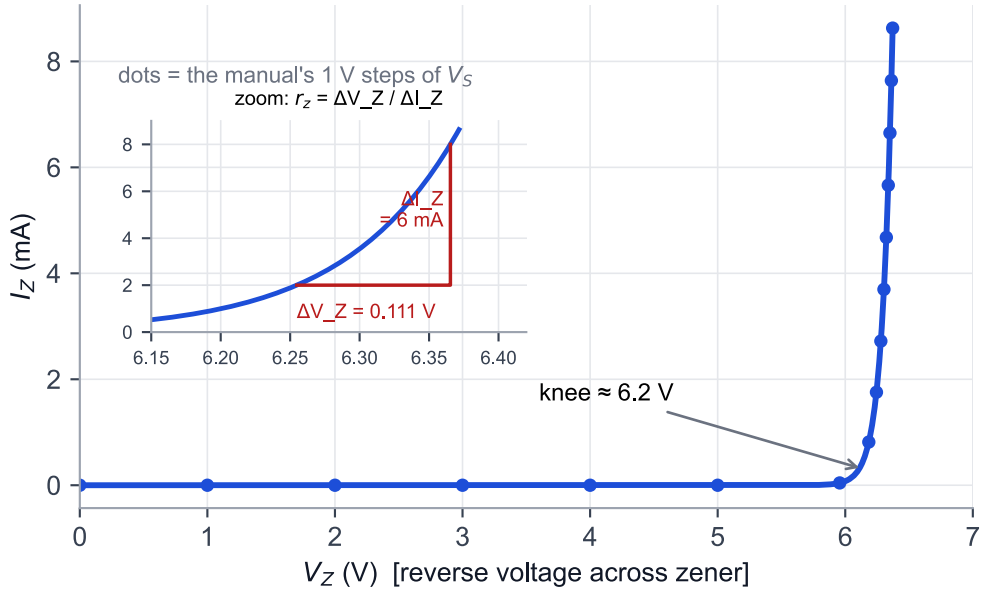
It lists a **1N4007**. That is an ordinary rectifier with a breakdown near 1000 V, so it will never break down on a 0–30 V supply. Ask for a **zener** (e.g. 6.2 V). A 1N4007 placed reverse-biased just shows µA leakage at every V_S.

DO IT (HANDS-ON, BREADBOARD)

1. Supply **off**, voltage knob at 0, current limit low.
2. Wire the 1 kΩ from supply + to the ammeter. Ammeter (DMM on mA DC, red lead in the mA port) to the zener's **cathode** (the banded end). Zener's anode to supply −.
3. Put the voltmeter (DMM on DC V, or the DSO) across the zener.
4. Switch on. Raise V_S in **1 V steps from 0 to 15 V**, reading V_Z and I_Z at each step.
5. Plot I_Z (y) against V_Z (x). Mark the knee: that is V_Z.
6. Take two points well into breakdown, read ΔV_Z and ΔI_Z, and calculate r_z.

What you should see

Simulated with a 6.2 V zener (BZX84C6V2L, the part Exp 2 uses). A different zener shifts the knee to its own V_Z, but the shape is identical.



LTspice, 1 kΩ series, V_S 0–15 V. Flat (open) until about 6 V, then nearly vertical. The zoom shows the r_z triangle.

V_S (V)	0–5	6	7	8	9	10	11	12	13	14	15
V_Z (V)	= V_S	5.957	6.185	6.245	6.279	6.303	6.322	6.337	6.350	6.362	6.372
I_Z (mA)	≈ 0 (µA)	0.04	0.82	1.76	2.72	3.70	4.68	5.66	6.65	7.64	8.63

Below breakdown, the zener takes almost no current, so no voltage drops across the 1 kΩ and $V_Z = V_S$. In breakdown, V_Z is pinned and the rest of V_S drops across the resistor: $I_Z = (V_S - V_Z)/1\ k\Omega$. Check: $V_S = 15$ gives $(15 - 6.372)/1000 = 8.63$ mA ✓.

Calculation

Using the 10 V and 15 V rows (both well into breakdown):

$$r_z = \frac{6.372 - 6.303}{8.63 - 3.70\ \text{mA}} = \frac{0.069\ \text{V}}{4.93\ \text{mA}} \approx 14\ \Omega$$

The zoom figure uses 2 → 8 mA and gives 0.111 V / 6 mA = 18 Ω. r_z shrinks as the current grows, so always **state the current range used**.

WRITE IN YOUR RECORD

Result: breakdown voltage $V_Z \approx$ **6.2 V** (at $I_Z \approx 1$ mA); dynamic resistance $r_z \approx$ **14 Ω** (between 3.7 and 8.6 mA). Use your own readings in the lab.

Theory for the record: a zener is a heavily doped pn diode designed to operate in reverse breakdown. Forward, it behaves like a normal diode (about 0.7 V). Reverse, only leakage flows until V_Z, where the current rises sharply while the voltage stays nearly constant. Breakdown is by tunnelling (below ~5 V) or avalanche (above ~6 V). Operation must stay between $I_{Z(min)}$ and $I_{Z(max)} = P_Z/V_Z$. The slope $\Delta V_Z/\Delta I_Z$ in breakdown is the dynamic resistance; a small value means good regulation, which Exp 2 uses.

TRAP: NO SERIES RESISTOR

Without the 1 kΩ (or with it shorted), the current in breakdown is limited only by the supply, and the zener burns out. Never connect a zener directly across a supply.

CHECK 1

(a) At $V_S = 12$ V, a student reads $V_Z = 6.34$ V. Predict I_Z without the ammeter. (b) Why is r_z not simply V_Z/I_Z? (c) A 3.3 V zener and a 12 V zener: which mechanism does each use, and which way does each one's V_Z move when heated?

Answers

Opening question. Reverse current rises sharply at V_Z because the junction breaks down (tunnelling or avalanche), producing many carriers. Dynamic resistance = $\Delta V_Z/\Delta I_Z$, the slope in breakdown. It is small (≈ 14–18 Ω here).

Check 1. (a) I_Z = (12 − 6.34)/1 kΩ = **5.66 mA**. (b) r_z describes how V changes when I changes (the slope), which is what matters for regulation. V_Z/I_Z at one point is a large, current-dependent number with no use here. (c) 3.3 V: Zener (tunnelling) mechanism, V_Z falls with heat (negative coefficient). 12 V: avalanche, V_Z rises with heat (positive coefficient).

Exp 2 · Zener diode as a voltage regulator: line and load regulation

Manual pp. 9–12 · LTspice / Virtual Lab · CO3 · the manual's printed data matches the simulation below digit-for-digit

What you already have: from Exp 1, in breakdown V_Z stays ≈ 6.2 V while I_Z changes. From C&S, voltage dividers and KCL.

Three terms first:

- **Regulator:** a circuit that keeps V_o constant.
- **Line regulation:** how much V_o moves when the **input** V_{in} changes. The "line" is the supply line.
- **Load regulation:** how much V_o moves when the **load current** changes, i.e. when R_L changes.

MANUAL EXP 2, AIM

To use a zener diode for line and load regulation. Viva form: "Input rises from 14 V to 17 V and V_o rises by only 0.08 V. Where did the other 2.92 V go?"

GUESS FIRST

Answer before reading on. Hint: there are only two other parts, R1 and the zener.

How a zener regulates

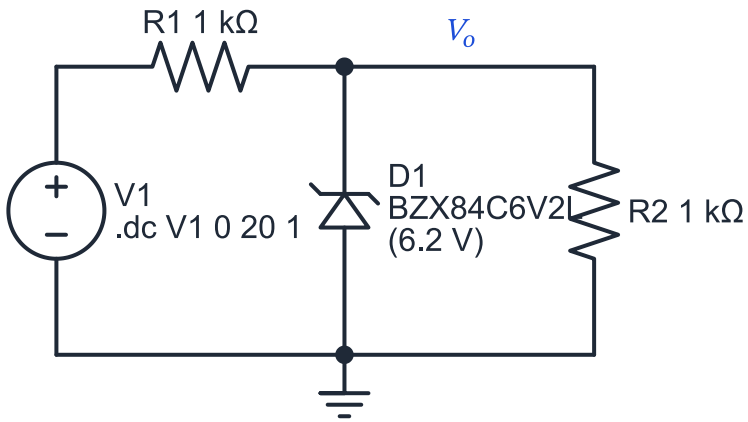


Fig 2.1 (line regulation): V1 swept 0→20 V, with a fixed 1 kΩ load.

KCL at the output node: the current through R1 splits between the zener and the load.

$$I_{R1} = \frac{V_{in} - V_Z}{R_1} = I_Z + I_L$$

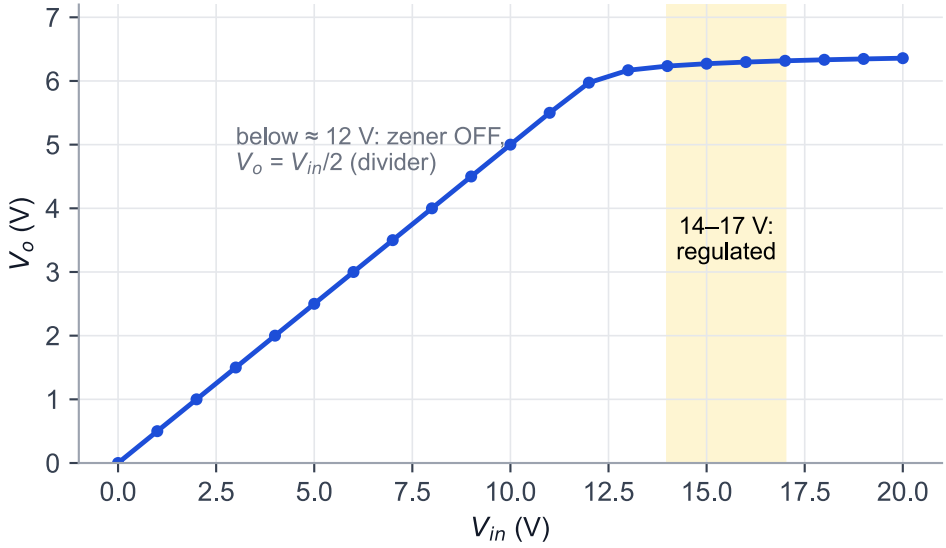
- **V_{in} rises:** the extra current all goes into the **zener**. V_Z barely moves (small r_z), so V_o stays put and **R1 absorbs the extra voltage**. That is the answer to the question.
- **R_L falls (more load current):** the zener gives up current to the load. V_o holds **as long as I_Z stays above $I_{Z(min)}$**.
- If the load demands more than R1 can supply, I_Z hits zero, the zener switches off, and V_o becomes an ordinary divider output that collapses.

Because the zener sits in parallel (shunt) with the load, this is called a **shunt regulator**.

Line regulation (Fig 2.1)

DO IT (LTSPICE)

1. Draw V1, R1 = 1k, zener **BZX84C6V2L** (cathode to the output node), R2 = 1k load, and ground.
2. Label the output wire `vo` (F4).
3. Directive: `.dc V1 0 20 1`. Run.
4. Plot V(vo) against V1. Use cursors to read V_o at V_{in} = 14 V and 17 V.



Two regions. Below about 12 V the zener is off and the output is half the input (1 kΩ:1 kΩ divider). Above it, V_o is pinned near 6.2–6.36 V.

Why 12 V? With the zener off, $V_o = V_{in}/2$. The zener turns on when $V_{in}/2$ reaches about 6.2 V, i.e. at $V_{in} \approx 12.4$ V.

V_{in} (V)	10	11	12	13	14	15	16	17	18	20
V_o (V)	5.000	5.500	5.973	6.168	**6.234**	6.271	6.297	**6.317**	6.333	6.358

$$\% \text{ line regulation} = \frac{\Delta V_o}{\Delta V_{in}} \times 100 = \frac{6.317 - 6.234}{17 - 14} \times 100 = \frac{0.083}{3} \times 100 = 2.8\,\%$$

Load regulation (Fig 2.2)

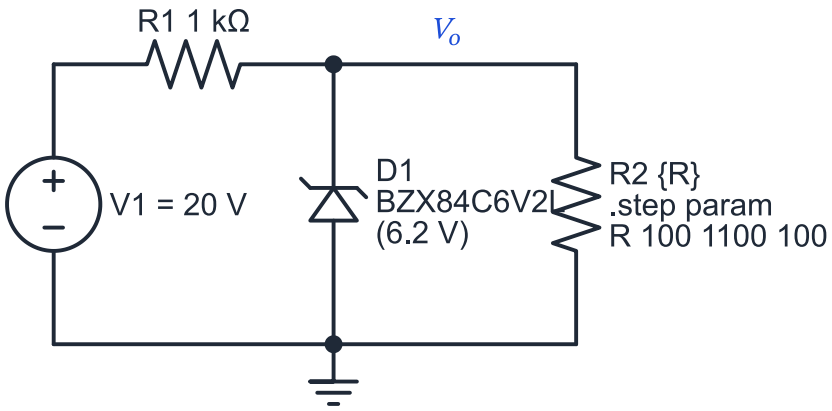
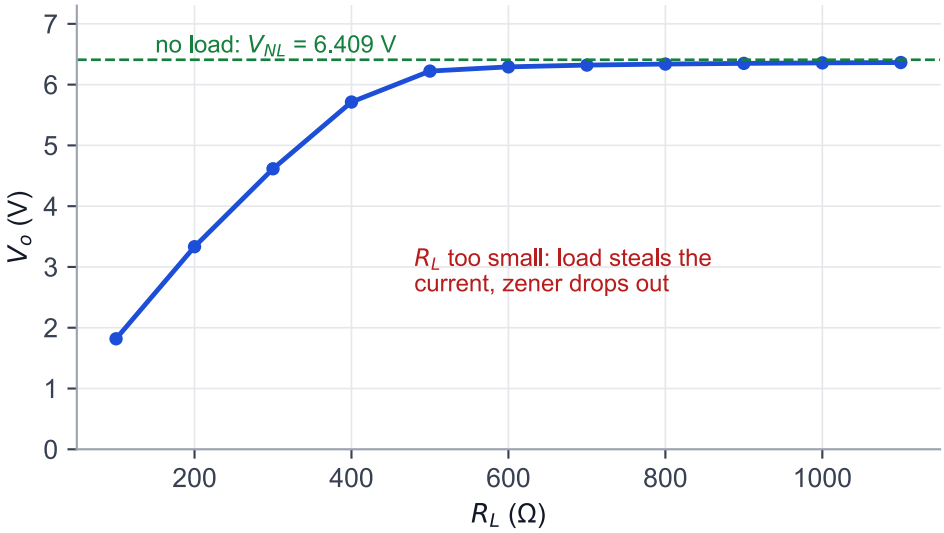


Fig 2.2 (load regulation): V1 fixed at 20 V, load stepped 100 Ω → 1100 Ω.

DO IT (LTSPICE)

1. Same circuit, but V1 = 20 V and the load value written as `{R}` (curly braces = a parameter).
2. Directives: `.step param R 100 1100 100` and `.op`. Run.
3. Plot V(vo). The x-axis is now R.
4. **No load:** delete R2, run `.op` again, and read V_{NL}.
5. Choose a full-load point (e.g. R = 1 kΩ) and read V_{FL}.



Above about 450 Ω the output holds near 6.3 V. Below that, the load takes more current than R1 can supply.

R_L (Ω)	100	200	300	400	500	600	800	1000	1100	no load
V_o (V)	1.818	3.333	4.615	5.714	6.223	6.292	6.338	**6.358**	6.364	**6.409**

$$\% \text{ load regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 = \frac{6.409 - 6.358}{6.358} \times 100 = 0.81\ \%$$

Smallest load that still regulates: R1 can pass at most (20 − 6.2)/1 kΩ = 13.8 mA. The load gets all of that only when $I_Z \to 0$, so $R_{L(min)}$ = 6.2 V / 13.8 mA ≈ **450 Ω**. This matches the knee in the plot. Below 450 Ω the zener is off, and $V_o = 20 \times R_L/(1000 + R_L)$: at 100 Ω that is 1.82 V ✓.

WRITE IN YOUR RECORD

Result: % line regulation = **2.8 %** (14–17 V, R_L = 1 kΩ); % load regulation = **0.81 %** (no load → 1 kΩ, V_{in} = 20 V). Lower % = better regulation.

Theory for the record: a zener shunt regulator places a reverse-biased zener across the load, fed through a series resistor R_S. In breakdown, the zener voltage is nearly constant, so changes in input voltage appear across R_S, and changes in load current are absorbed by a change in zener current. Regulation holds while $I_{Z(min)} < I_Z < I_{Z(max)}$. R_S is chosen so that $I_Z \geq I_{Z(min)}$ at minimum input and maximum load. Line regulation = $\Delta V_o/\Delta V_{in} \times 100$ %; load regulation = $(V_{NL} - V_{FL})/V_{FL} \times 100$ %.

TRAP: AN "UNREGULATED" LINE PLOT

If V_o keeps rising with slope ½ all the way to 20 V, the zener is in backwards (forward-biased, clamping at about 0.7 V) or is the wrong part. If V_o sticks at about 0.7 V, it is definitely reversed. The cathode (bar) must face the output node.

CHECK 1

(a) Why does V_o in the line plot still creep up slightly (6.23 → 6.36 V) instead of staying perfectly flat? (b) With V_{in} = 20 V and R_L = 1 kΩ, how much current does the zener carry? (c) What happens to I_Z when the load is removed, and why is that the worst case for zener power?

Answers

Opening question. Across R1. The extra input raises the current through R1 (it flows on into the zener), so R1's voltage drop rises by about 2.92 V.

Check 1. (a) The zener has a dynamic resistance r_z (Exp 1). More current through it means slightly more voltage: $\Delta V = r_z \Delta I_Z$. (b) $I_{R1} = (20 - 6.358)/1000 = 13.64$ mA, $I_L = 6.358/1000 = 6.36$ mA, so $I_Z = 13.64 - 6.36 =$ **7.28 mA**. (c) All of R1's current then flows through the zener, its maximum, so I_Z and $P_Z = V_Z I_Z$ peak. Design must keep this below $I_{Z(max)}$.

Exp 3 · Diode clipper circuits

Manual pp. 13–14 · hands-on · CO3

0 Instruments → 1 Zener curve → 2 Regulator → **3 Clipper** → 4 Clamper

What you already have: a silicon diode conducts once its anode is about 0.7 V above its cathode (Exp 1, forward region). Below that it is an open circuit.

Three terms first:

- **Clipper:** a circuit that cuts off the part of a waveform above (or below) a set level.
- **Clipping level:** the voltage at which the cut happens.
- **Cut-in voltage V_γ:** the forward voltage at which a diode starts to conduct (≈ 0.7 V for silicon).

MANUAL EXP 3, AIM

To implement a diode clipper circuit and observe its output waveform. Viva form: "A 10 V-peak sine feeds Fig 3.1(a). At what level is it clipped, and which half is affected?"

GUESS FIRST

Look at the circuit below and commit to a number and a half (top or bottom) before reading on.

How it works

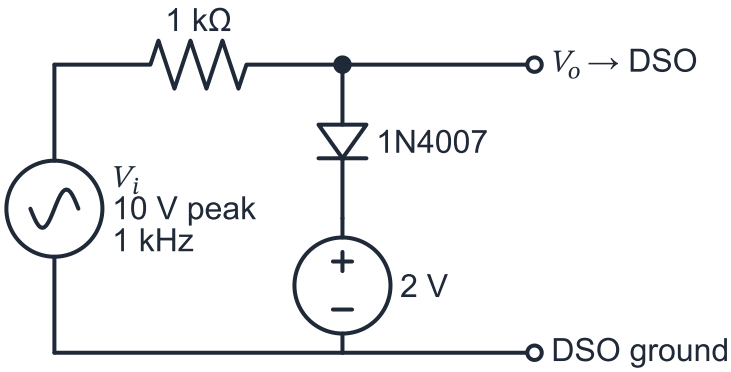


Fig 3.1(a): diode anode at V_o, cathode at the + of a 2 V battery.

Treat the diode as a switch, and ask when its anode is more than 0.7 V above its cathode.

- The cathode sits at **+2 V** (the battery).
- The diode conducts when $V_o > 2 + 0.7 =$ **2.7 V**. Then V_o is held at 2.7 V, and the rest of V_{in} drops across the 1 kΩ.
- When $V_{in} < 2.7$ V the diode is off: no current flows, so there is no drop across the 1 kΩ, and $\mathbf{V_o = V_{in}}$.

Result: the **top** of the sine is cut flat at +2.7 V, and the bottom passes untouched.

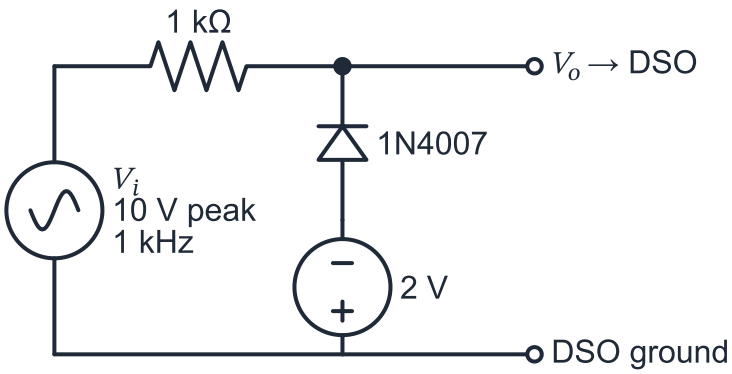
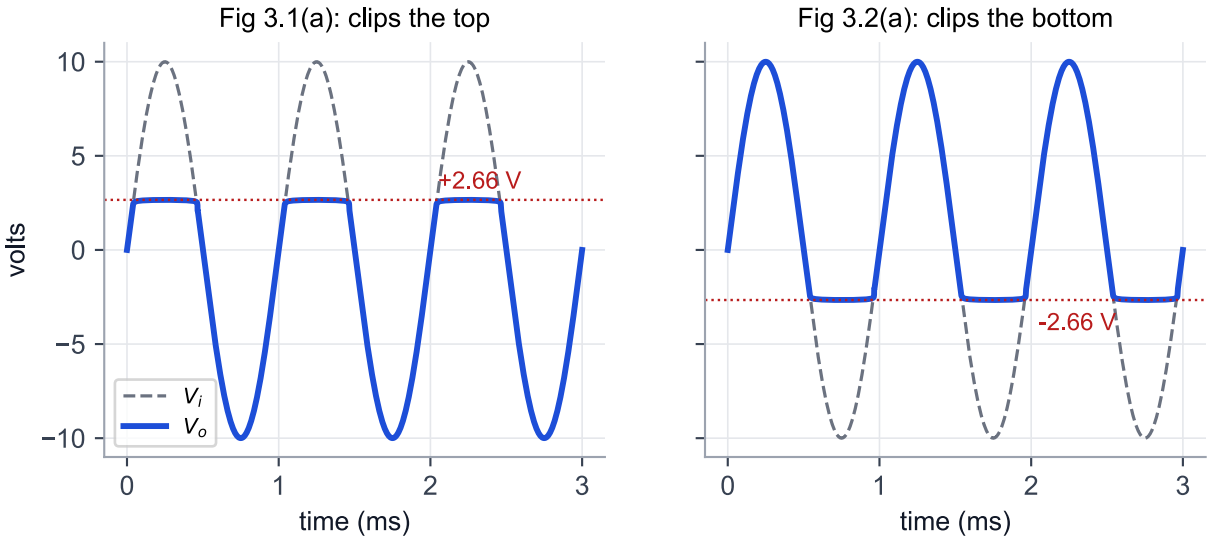


Fig 3.2(a): diode and battery both reversed.

Here the diode's anode is at −2 V. It conducts when $V_o < -2.7$ V, so the **bottom** is clipped at −2.7 V.



LTspice with a 1N4007: clipping at ±2.66 V (this diode drops 0.66 V at the clipping current).

Transfer characteristic (V_o plotted against V_i, manual Fig 3.3): a 45° line ($V_o = V_i$) that turns flat at +2.7 V for (a), or at −2.7 V for (b).

DO IT (HANDS-ON)

1. Generator: sine, **10 V peak** (20 V p-p on the DSO), 1 kHz. Check it on the DSO **before** connecting the circuit.
2. Breadboard: generator → 1 kΩ → output node. From the output node, connect the diode with its **anode at the output** (band/cathode away from the output), then its cathode to the **+** of the 2 V supply. Supply **−** goes to the generator's ground.
3. DSO CH1 on V_{in} and CH2 on V_o, both grounds at circuit ground. Coupling **DC** (AC coupling shifts the trace and hides the true level).
4. Read the clipping level: divisions × VOLTS/DIV.
5. Swap the diode **and** the supply polarity to build Fig 3.2(a). Repeat.
6. Optional: DSO **X-Y mode** (CH1 = x, CH2 = y) draws the transfer characteristic directly.

WRITE IN YOUR RECORD

Result: clipping level ≈ **+2.7 V** for Fig 3.1(a) (positive clipper) and **−2.7 V** for Fig 3.2(a) (negative clipper). Measured with the 1N4007: about ±2.66 V.

Theory for the record: a clipper removes the part of a signal beyond a chosen level without disturbing the rest. A diode in series with a DC reference V_R, placed across the output, conducts when the output exceeds $V_R + V_\gamma$, holding the output at that level. Otherwise the diode is off and $V_o = V_i$. Reversing the diode and the reference clips the opposite half. The series resistor limits the current and drops the excess voltage. Clippers are used for wave shaping and to protect inputs from overvoltage.

TRAP: THE DSO ON AC COUPLING

AC coupling removes the DC average, so the flat top no longer reads +2.7 V. Use DC coupling for any clipping-level or clamping measurement.

CHECK 1

(a) Change the battery in Fig 3.1(a) to 5 V. New clipping level? (b) Remove the battery (short it) in Fig 3.1(a). New clipping level? (c) Why must the series 1 kΩ be there at all?

Answers

Opening question. At about +2.7 V (2 V battery + 0.7 V diode). The **positive** half is clipped.

Check 1. (a) 5 + 0.7 = **+5.7 V**. (b) **+0.7 V**: a plain diode clipper. (c) While the diode conducts, the output is pinned. The difference V_{in} − 2.7 V must drop somewhere, and the resistor takes it. Without it, the generator would drive a large current straight into the diode and battery.

Exp 4 · Diode clamper circuit

Manual pp. 15–16 · LTspice · CO3 · the manual's sample output minimum (−9.38 V) matches the simulation below

0 Instruments → 1 Zener curve → 2 Regulator → 3 Clipper → **4 Clamper**

What you already have: the clipper from Exp 3, which **cuts** a wave. From C&S, a capacitor keeps its charge unless current flows, and τ = RC sets how fast it discharges.

Two terms first:

- **Clamper (DC restorer):** a circuit that **shifts** the whole wave up or down without changing its shape. It adds a DC level.
- **Peak-to-peak preserved:** the wave keeps its full V_{pp}. Only its position moves.

MANUAL EXP 4, AIM

To implement a diode clamper circuit and observe its output waveform. Viva form: "Input is ±5 V. The output swings from about +0.5 V to −9.4 V. Explain where the −4.4 V shift comes from."

GUESS FIRST

Which part could be storing a voltage that adds to the input?

How it works

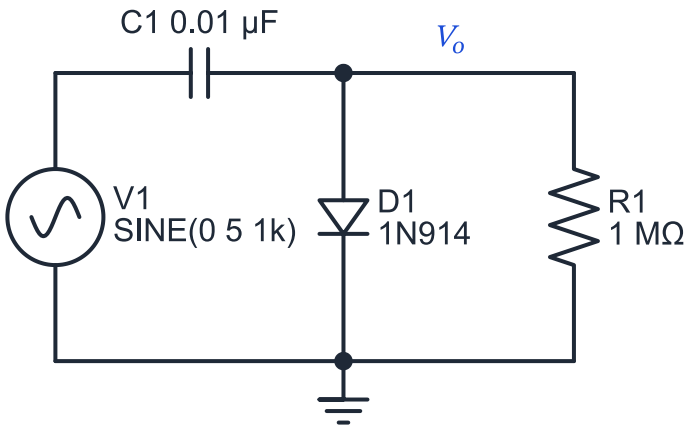
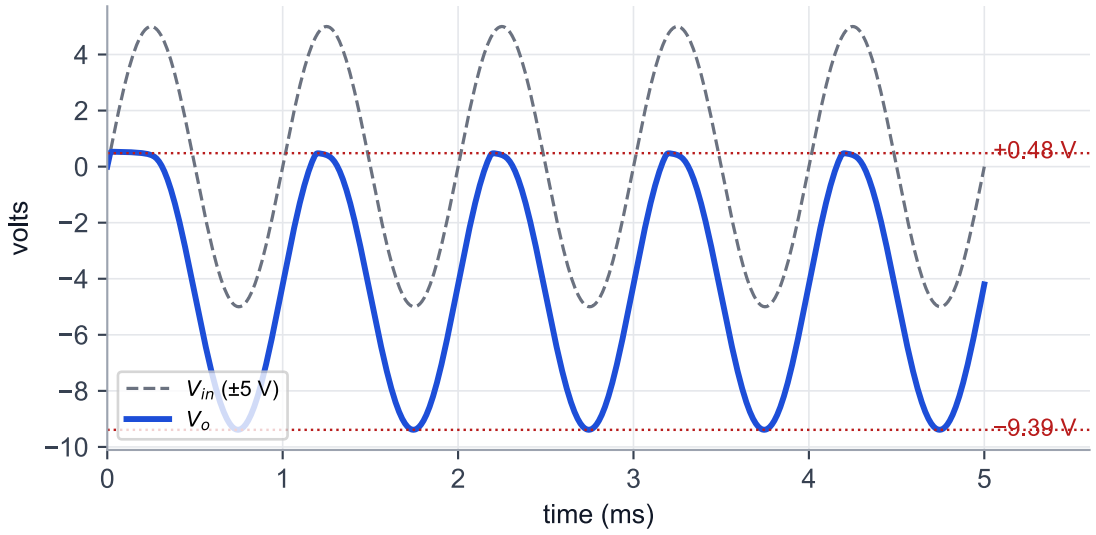


Fig 4.1: series C, diode to ground (anode at V_o), 1 MΩ load.

1. **First positive peak:** V_{in} rises, V_o tries to go positive, and the diode turns on, holding $V_o \approx +0.5$ V. The capacitor charges to about $\mathbf{V_p - 0.5 \approx 4.5\ V}$, positive on the input side.
2. **After that** the diode is off. The capacitor cannot discharge quickly, because its only path is the 1 MΩ: τ = RC = 1 MΩ × 0.01 µF = **10 ms**, which is 10× the 1 ms period. So it acts like a 4.5 V battery in series with the input:

$$V_o = V_{in} - V_C \approx V_{in} - 4.5\ V$$

So the output swings from +5 − 4.5 = **+0.5 V** to −5 − 4.5 = **−9.5 V**. Same 10 V p-p, but shifted down by the capacitor's voltage. This is a **negative clamper**: the positive peak is clamped at about 0 V, and the wave sits below it.



LTspice: V_o from +0.48 V to −9.39 V. The output is clamped in the very first cycle.

Reverse the diode (cathode at V_o): the negative peak clamps at about −0.5 V, and the wave sits above it, a **positive clamper**. **Add a battery** in series with the diode: the clamping level moves to that battery's voltage ± 0.7 V.

DO IT (LTSPICE)

1. V1: right-click, **Advanced** → **SINE**, DC offset 0, amplitude 5, frequency 1k (= `SINE(0 5 1k)`).
2. C1 = `0.01u`. Diode D1: Pick New Diode → **1N914**, with its anode on the output node and cathode to ground. R1 = `1Meg` (**not** `1M`!).
3. Directive: `.tran 0 5m 0 2u`. Run. Plot the input wire and V(vo).
4. Use cursors to read the maximum and minimum of V_o. Then V_{pp} = max − min, and the DC shift = (max + min)/2.

Quantity	LTspice value
V_o maximum (clamped level)	+0.48 V
V_o minimum	−9.39 V
V_o peak-to-peak	9.87 V (input 10 V)
average of V_o (DC shift)	−4.41 V

WRITE IN YOUR RECORD

Result: output clamped at about **+0.5 V** (positive peak); V_o swings +0.48 V to −9.39 V; peak-to-peak 9.87 V ≈ input 10 V; DC level shifted by about −4.4 V. Negative clamper.

Theory for the record: a clamper shifts a waveform's DC level without changing its shape or peak-to-peak value. A capacitor in series charges through the diode to (nearly) the input peak on the first half cycle. With the diode off, it discharges only through the load, so it must have RC ≫ T (here 10 ms ≫ 1 ms). The capacitor voltage then adds to the input, shifting the wave so that one peak sits at the diode's clamping level. Diode direction sets negative/positive clamping; a series battery sets the reference level.

TRAPS

- `1M` in LTspice is 1 **milli**ohm. With R ≈ 0, the capacitor discharges every cycle and nothing clamps.
- If RC is comparable to T, the capacitor droops between peaks and the output tilts. The rule is RC ≥ 10T (some books use 5T).

CHECK 1

(a) Clipper vs clamper: one sentence each on what happens to V_{pp}. (b) Replace 0.01 µF with 100 pF. What goes wrong, and why? (c) Diode reversed: give V_o's max and min.

Answers

Opening question. The capacitor charges to about 4.5 V on the first positive peak (through the diode). It then can't discharge (τ = 10 ms $\gg$ 1 ms), so it stays in series with the input as a −4.5 V offset.

Check 1. (a) A clipper **reduces** V_{pp} (part of the wave is cut). A clamper **keeps** V_{pp} and only shifts the DC level. (b) τ = 1 MΩ × 100 pF = 0.1 ms $\ll$ 1 ms: the capacitor discharges between peaks, so the shift is not held and the output distorts. (c) Positive clamper: min ≈ −0.5 V, max ≈ +9.5 V.

Exp 5 · BJT input/output characteristics (CE) and h-parameters

Manual pp. 17–19 · LTspice / Virtual Lab · CO1, CO2 · theory: ECE2101 unit 5 (BJTs)

5 BJT curves → 6 CE amplifier → 7 Emitter follower → 8 Feedback

What you already have: a diode is one pn junction (Exp 1). It conducts forward above ≈ 0.7 V.

Three terms first:

- **BJT (bipolar junction transistor):** three layers, **n-p-n** for an NPN. The layers are the **emitter** (E, heavily doped), the **base** (B, very thin, lightly doped) and the **collector** (C). That makes two junctions: base–emitter (BE) and base–collector (BC).
- **Common emitter (CE):** the emitter is shared by the input (B–E) and the output (C–E).
- **h-parameters:** four numbers that describe how the transistor responds to small changes. Exp 6–8 amplifiers are built on them.

MANUAL EXP 5, AIM

To plot the input and output characteristics of an NPN transistor in CE configuration and determine h-parameters from the characteristics. Viva form: "Your output curves are flat lines spaced 5.5 mA apart for 20 µA steps of base current. What is h_{fe}, and what does a flat line tell you?"

GUESS FIRST

Commit to a number for h_{fe}, and to a one-word meaning of "flat", before reading on.

The BJT in five lines

1. Forward-bias the BE junction (V_{BE} ≈ 0.6–0.7 V). The emitter injects electrons into the base.
2. The base is thin, so almost all of them cross it and are swept into the collector by the reverse-biased BC junction. A few recombine in the base: that is the base current.
3. So a **small base current controls a large collector current**: $I_C = \beta I_B$, with β ≈ 100–300 (≈ 290 for the BC547B).
4. KCL: $\mathbf{I_E = I_B + I_C}$.
5. Which region the transistor is in depends on how the two junctions are biased:

Region	BE junction	BC junction	Behaviour	Used as
Active	forward	reverse	$I_C = \beta I_B$, nearly independent of V_{CE}	amplifier (Exps 6–8, 11)
Saturation	forward	forward	V_{CE} ≈ 0.2 V, $I_C < \beta I_B$	closed switch
Cut-off	reverse	reverse	I_C ≈ 0	open switch

The two families of curves

- **Input characteristic:** I_B against V_{BE}, at fixed V_{CE}. It is just the BE diode's forward curve.
- **Output characteristic:** I_C against V_{CE}, at fixed I_B. One curve per I_B.

h-parameters in CE form (small changes, written in lowercase):

$$v_{be} = h_{ie}\, i_b + h_{re}\, v_{ce} \qquad i_c = h_{fe}\, i_b + h_{oe}\, v_{ce}$$

Parameter	Name	Read from	Formula
h_{ie} (= h_{11})	input resistance	input curve, V_{CE} fixed	$\Delta V_{BE}/\Delta I_B$
h_{re} (= h_{12})	reverse voltage gain	input curves, I_B fixed	$\Delta V_{BE}/\Delta V_{CE}$
h_{fe} (= h_{21})	forward current gain	output curves, V_{CE} fixed	$\Delta I_C/\Delta I_B$
h_{oe} (= h_{22})	output admittance	one output curve, I_B fixed	$\Delta I_C/\Delta V_{CE}$

The manual swaps the names for h_{21} and h_{22}, and writes "$1/h_{oe} = \Delta I_C/\Delta V_{CE}$". The table above is the correct version. $\Delta I/\Delta V$ is a conductance, so h_{oe} is in siemens and $1/h_{oe}$ is the output resistance.

The circuit, and one fix

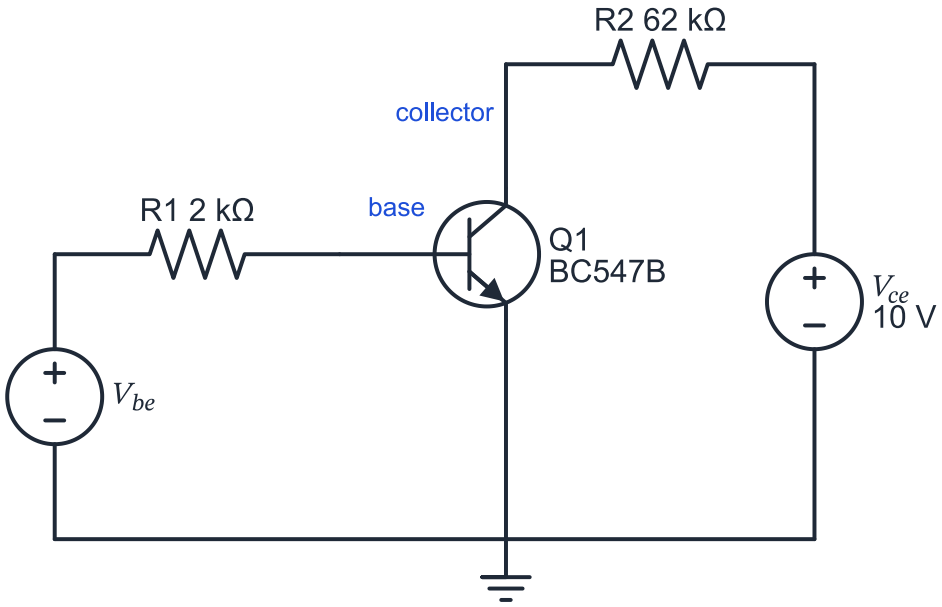


Fig 5.1 as printed: V_{be} through 2 kΩ to the base; V_{ce} through 62 kΩ to the collector.

TRAP: THE 62 KΩ MAKES THE MANUAL'S TABLE IMPOSSIBLE

With 62 kΩ in the collector, $I_C \le 10\ V / 62\ k\Omega =$ **0.16 mA**. The transistor saturates as soon as I_B passes about 0.6 µA, so "V_{CE}" is no longer the 10 V you set, and the I_B = 20/40 µA curves of the observation table can never appear. That is why the manual's own output plot only reaches about 210 µA. The fix: **replace R2 with a wire** so V_{ce} sits directly on the collector, and drive the base with a **current source** for the output curves. The device and the directives stay the same.

DO IT (LTSPICE)

Input characteristics:

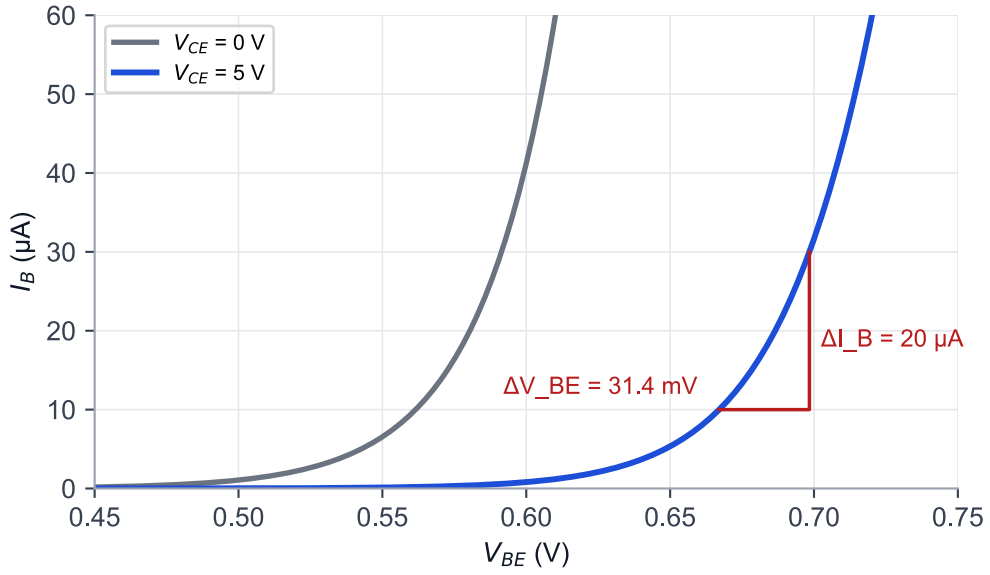
1. Draw Fig 5.1 with R2 replaced by a wire. Q1: Pick New Transistor → **BC547B**. Label the base node `base`.
2. Directive: `.dc Vbe 0 1 0.001 Vce 0 5 5` (V_{CE} = 0 V and 5 V, as the observation table asks).
3. Run, click the base lead to plot **Ib(Q1)**, then right-click the x-axis and set it to **V(base)**.

Output characteristics:

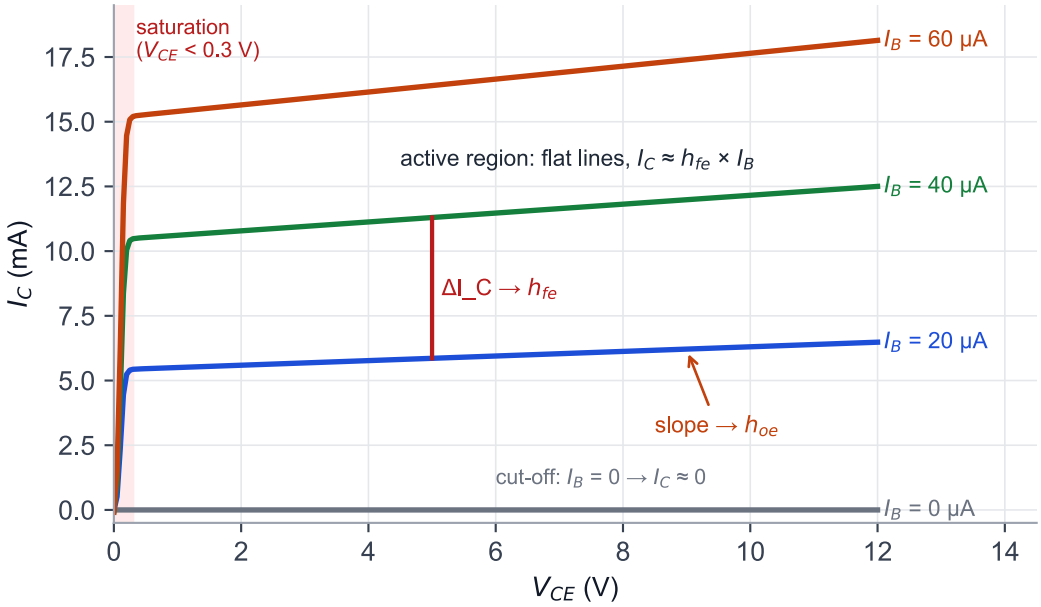
1. Replace V_{be} and R1 with a **current source** `Ib` (arrow pointing **into** the base).
2. Directive: `.dc Vce 0 12 0.05 Ib 0 60u 20u` (I_B = 0, 20, 40, 60 µA).
3. Plot **Ic(Q1)** against V_{ce}.

h_{re}: set Ib = `20u`, run `.op` at V_{ce} = 2 V and at 10 V, and read V(base) both times.

What you should see



Input characteristic: a diode curve. Raising V_{CE} shifts it right (at V_{CE} = 0 the BC junction is also forward and adds base current).



Output characteristics: saturation hugging the left edge, the active region (nearly flat), and cut-off at $I_B = 0$.

V_{CE} (V)	0.5	1	2	4	6	8	10	12
I_C (mA), I_B = 0	0	0	0	0	0	0	0	0
I_C (mA), I_B = 20 µA	5.46	5.50	5.59	**5.77**	5.95	6.13	**6.30**	6.48
I_C (mA), I_B = 40 µA	10.53	10.61	10.78	11.13	11.47	11.81	12.16	12.50
I_C (mA), I_B = 60 µA	15.27	15.40	15.65	16.15	16.65	17.14	17.64	18.14

Input, V_{CE} = 5 V: V_{BE} = 0.60 V → 0.81 µA; 0.62 → 1.73; 0.64 → 3.68; 0.66 → 7.73; 0.68 → 15.9; 0.70 → 31.6 µA. At V_{CE} = 0 V: 0.50 V → 1.07 µA; 0.55 → 6.55; 0.60 → 41.2 µA.

Calculations (from the curves above)

$$h_{ie} = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{0.6984 - 0.6670}{30 - 10\ \mu A} = \frac{31.4\ mV}{20\ \mu A} = 1.57\ k\Omega \qquad (V_{CE} = 5\ V)$$

$$h_{fe} = \frac{\Delta I_C}{\Delta I_B} = \frac{11.30 - 5.86\ mA}{40 - 20\ \mu A} = \frac{5.44\ mA}{20\ \mu A} = 272 \qquad (V_{CE} = 5\ V)$$

$$h_{oe} = \frac{\Delta I_C}{\Delta V_{CE}} = \frac{6.30 - 5.77\ mA}{10 - 4\ V} = \frac{0.533\ mA}{6\ V} = 89\ \mu S \qquad (I_B = 20\ \mu A;\ 1/h_{oe} = 11.2\ k\Omega)$$

$$h_{re} = \frac{\Delta V_{BE}}{\Delta V_{CE}} = \frac{0.68671 - 0.68638}{10 - 2} = \frac{0.33\ mV}{8\ V} = 4.2 \times 10^{-5} \qquad (I_B = 20\ \mu A)$$

Sense check: h_{re} is tiny (the output barely affects the input), and h_{oe} is small (the flat lines tilt only slightly). That is why simple amplifier analysis often drops both.

WRITE IN YOUR RECORD

Result (BC547B, CE): $h_{ie} \approx$ **1.57 kΩ**, $h_{re} \approx$ **4.2×10^{-5}**, $h_{oe} \approx$ **89 µS**, $h_{fe} \approx$ **272**. State the operating point used: V_{CE} = 5 V, $I_B \approx$ 20 µA.

Theory for the record: a BJT has emitter, base and collector regions forming two junctions. For amplification the BE junction is forward-biased and the BC junction reverse-biased (active region): carriers injected from the emitter cross the thin base and reach the collector, so $I_C = \beta I_B$ and $I_E = I_B + I_C$. In CE configuration, the input characteristic (I_B vs V_{BE} at constant V_{CE}) resembles a forward diode curve. The output characteristics (I_C vs V_{CE} at constant I_B) show saturation, active and cut-off regions. The h-parameters h_{ie}, h_{re}, h_{fe}, h_{oe} are the slopes of these curves at the operating point and form the small-signal model.

CHECK 1

(a) Using the table, find h_{fe} at V_{CE} = 10 V between I_B = 40 and 60 µA. (b) Why is the output curve for I_B = 0 flat at zero? (c) Which region is the transistor in at V_{CE} = 0.1 V and I_B = 40 µA, and how can you tell?

Answers

Opening question. h_{fe} = 5.5 mA / 20 µA ≈ **275**. "Flat" means **I_C does not depend on V_{CE}**: the transistor acts as a current source controlled by I_B. That is the active region, and the slight tilt is h_{oe}.

Check 1. (a) (17.64 − 12.16) mA / 20 µA = 5.48 mA / 20 µA = **274**. (b) With no base current, the BE junction isn't injecting carriers, so the transistor is cut off and only a leakage current (nA) flows. (c) **Saturation.** V_{CE} is below about 0.2–0.3 V, the steep left-hand part of the curve, where I_C is less than βI_B and depends strongly on V_{CE}.

Exp 6 · CE amplifier: frequency response and bandwidth

Manual pp. 20–21 · hands-on · CO4, CO5 · the same stage is reused in the extra experiment

5 BJT curves → **6 CE amplifier** → 7 Emitter follower → 8 Feedback

What you already have: from Exp 5, in the active region $I_C = \beta I_B$, and the BE junction needs ≈ 0.7 V. From C&S, a capacitor's reactance $X_C = 1/(2\pi fC)$: huge at low f, tiny at high f.

Three terms first:

- **Biasing:** DC voltages that park the transistor in the active region at a chosen **Q-point** (quiescent point = the DC operating point with no signal).
- **Small-signal gain A_v:** v_o/v_{in} for a small sine riding on the Q-point. In decibels: $20 \log_{10} A_v$.
- **Bandwidth:** the frequency range over which the gain stays within 3 dB (a factor of 0.707) of its maximum.

MANUAL EXP 6, AIM

To plot the frequency response of a CE amplifier and determine its bandwidth. Viva form: "Why does the gain fall at **both** low and high frequencies, and what sets each cut-off?"

GUESS FIRST

Name one part responsible for each end before reading on.

The circuit, part by part

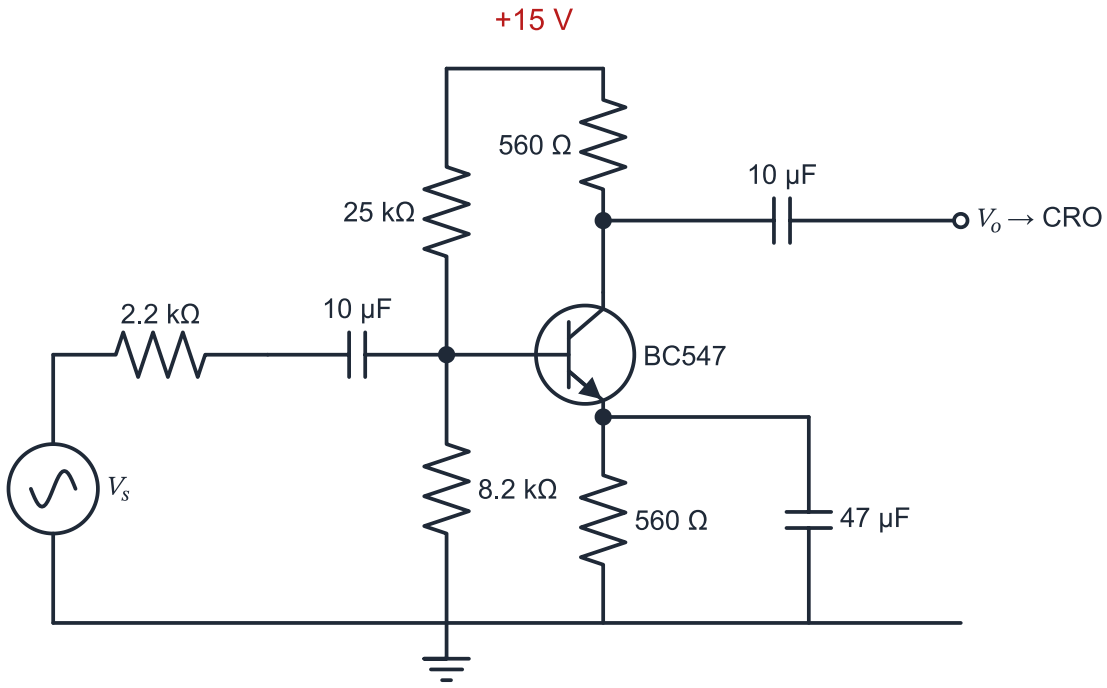


Fig 6.1, redrawn: voltage-divider bias, emitter resistor with bypass capacitor, coupling capacitors in and out.

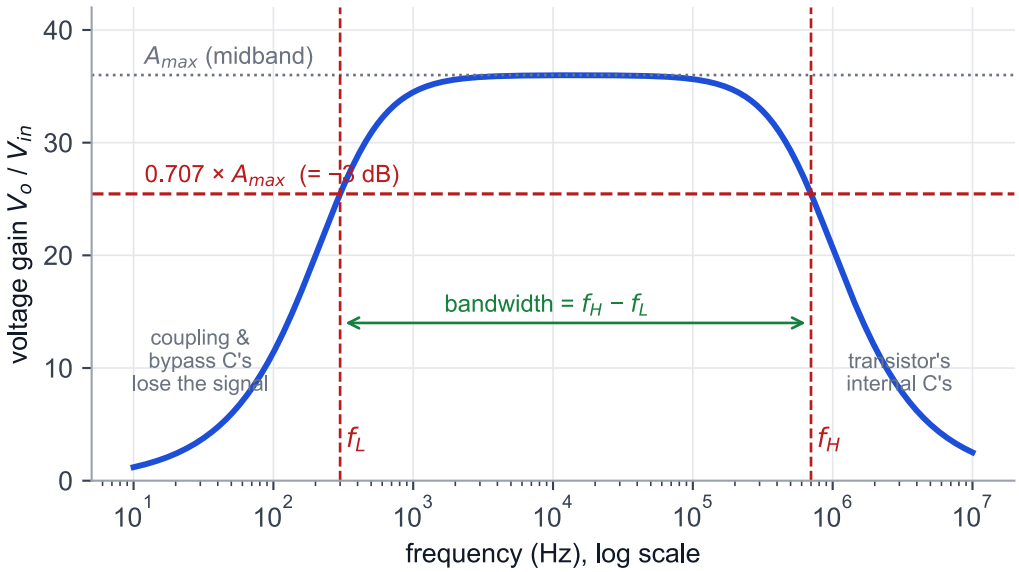
Part	Job
25 kΩ + 8.2 kΩ (R1, R2)	voltage divider: fixes $V_B \approx 15 \times 8.2/33.2 \approx 3.6$–$3.7$ V
560 Ω emitter (R_E)	sets $I_E = V_E/R_E$. It also stabilises the bias: if I_C rises with temperature, V_E rises, which reduces V_{BE}.
47 µF across R_E (C_E, **bypass**)	shorts R_E for AC, so the signal sees the emitter grounded. This gives full gain.
10 µF input (C_{in}, **coupling**)	passes the signal, blocks DC (so the generator doesn't upset the bias)
10 µF output (C_C)	passes the amplified signal, blocks the 12 V collector DC from the CRO
560 Ω collector (R_C)	turns the collector current change into a voltage change

Q-point (LTspice .op): $V_B = 3.60$ V, $V_E = 2.92$ V, $I_C = 5.2$ mA, $V_C = 12.09$ V, so $V_{CE} = 9.2$ V. The transistor is safely active.

Gain by hand:

- r_e = 26 mV / I_E = 26/5.2 = **5 Ω**. This is the BE junction's small-signal resistance.
- Base to collector: $A_v = -R_C/r_e = -560/5 = -112$.
- The 2.2 kΩ source resistor and the amplifier's input resistance (R_{in} = 25k ∥ 8.2k ∥ βr_e ≈ 1.2 kΩ) form a divider. Only 1.2/(1.2 + 2.2) = 0.35 of v_s reaches the base.
- Overall: $v_o/v_s \approx 0.35 \times 112 \approx$ **39**. LTspice gives **36**.
- The minus sign means the output is **inverted** (180° phase shift).

Why the gain falls at both ends



The general shape. Low end: capacitor reactances grow. High end: the transistor's internal junction capacitances short the signal.

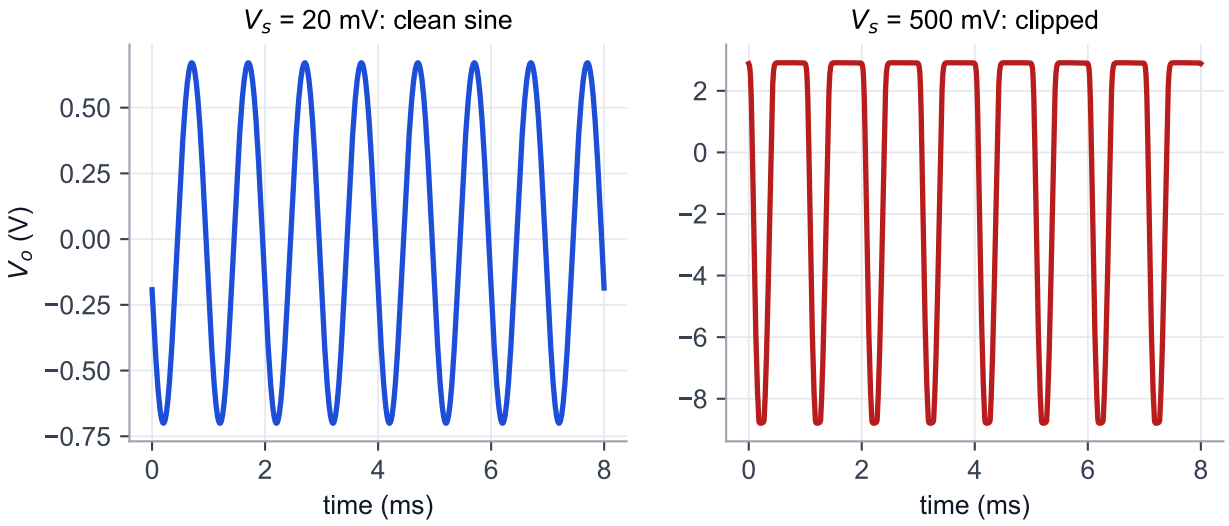
- **Low f:** $X_C = 1/(2\pi fC)$ grows. The coupling capacitors start to block the signal, and above all **C_E stops bypassing R_E**, which brings in negative feedback (see Exp 8). The lower cut-off is set mostly by C_E: $f_L \approx 1/(2\pi\, C_E\, R_e')$, where $R_e' \approx r_e$ + (source ∥ bias)/β ≈ 10.6 Ω. That gives 1/(2π × 47 µF × 10.6 Ω) ≈ **320 Hz**.
- **High f:** tiny capacitances inside the transistor (between its junctions) and in the wiring start to short the signal. The base–collector capacitance is multiplied by the gain (the *Miller effect*), so it dominates.
- **Middle (midband):** all external capacitors act as shorts and the internal ones as opens, so the gain is flat.

DO IT (HANDS-ON)

1. Build Fig 6.1 on the breadboard with supply **+15 V**. 25 kΩ is not a standard value: use 22 kΩ + 3.3 kΩ in series. Check the electrolytic capacitors' polarity: + towards the higher DC voltage (base side for C_{in}, collector side for C_C, emitter side for C_E).
2. **Check the Q-point first** with the DMM, no signal applied: $V_B \approx 3.6$ V, $V_E \approx 2.9$ V, $V_C \approx 12$ V. If $V_C \approx 15$ V the transistor is off; if $V_C \approx V_E$ it is saturated. Fix this before going on.
3. Generator: sine, 1 kHz. Set the amplitude **small: about 20 mV peak** (use the generator's −20 dB/attenuator button, or a 1 kΩ:100 Ω divider). Look at V_o on the DSO. It must be a clean sine, not flattened.
4. Keep V_s constant. Step the frequency through **100 Hz, 500, 800, 1k, 5k, 10k, 50k, 100k, 500k, 800k, 1 MHz** and read $V_{o(pp)}$ at each.
5. Calculate $A_v = V_o/V_s$ and 20 log A_v. Plot dB (y) against frequency on **semi-log** paper (log x-axis).
6. Draw a line 3 dB below the maximum. Where it cuts the curve gives f_L and f_H. BW = $f_H - f_L$.

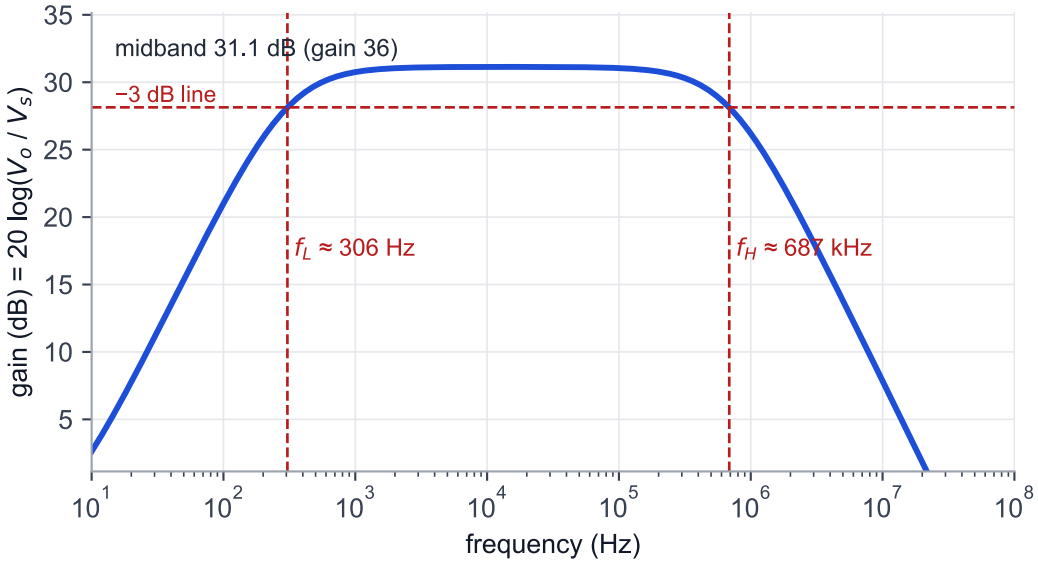
TRAP: THE MANUAL'S 500 MV INPUT

At 500 mV the output would need to swing about ±18 V, but the collector can only move between ≈ 3 V and 15 V. The output is clipped flat (below), so the "gain" you read is meaningless. Even the 100 mV in Fig 6.1 distorts (+2.5 / −3.4 V). Use about 20 mV.



LTspice: the same amplifier with 20 mV in (clean, gain ≈ 34) and 500 mV in (clipped).

What you should see



LTspice, Fig 6.1 with the CRO as a 1 MΩ load. Midband gain 36 (31.1 dB), $f_L \approx 306$ Hz, $f_H \approx 687$ kHz.

f (Hz)	100	500	800	1k	5k	10k	50k	100k	500k	800k	1M
$A_v = V_o/V_s$	11.2	30.8	33.7	34.5	36.0	36.1	36.0	35.7	29.2	23.5	20.4
gain (dB)	21.0	29.8	30.6	30.8	31.1	31.1	31.1	31.1	29.3	27.4	26.2

On the bench, f_L should land close to 300 Hz, because it is set by the capacitors you placed. f_H depends on breadboard wiring, probe and generator capacitance, so expect anything from a few hundred kHz upward. The simulation is the ideal-wiring value.

Calculation

$$BW = f_H - f_L = 687\ \text{kHz} - 0.306\ \text{kHz} \approx 687\ \text{kHz}$$

When $f_H \gg f_L$, the bandwidth is essentially f_H.

WRITE IN YOUR RECORD

Result: maximum gain ≈ **36 (31.1 dB)**; $f_L \approx$ **306 Hz**, $f_H \approx$ **687 kHz**; bandwidth ≈ **687 kHz** (simulation; use your measured values).

Theory for the record: in a CE amplifier the input is applied at the base and the output taken at the collector, with the emitter common (grounded for AC by the bypass capacitor). Voltage-divider bias with an emitter resistor sets a stable Q-point in the active region. A small base signal produces a β-times-larger collector current change, which develops a large, phase-inverted voltage across R_C. Gain ≈ R_C/r_e. The frequency response is flat in midband, falls at low frequencies because of the coupling and bypass capacitors, and falls at high frequencies because of junction and stray capacitances. Bandwidth = $f_H - f_L$, measured at the −3 dB (0.707 A_{max}) points, also called the half-power points.

CHECK 1

(a) Why is the −3 dB point called the "half-power" point? (b) You remove C_E. Does the gain go up or down, and roughly to what? (c) V_C reads 15 V with no signal. What is wrong?

Answers

Opening question. Low end: the **coupling and bypass capacitors**' reactance becomes large; C_E matters most ($f_L \approx 1/(2\pi C_E R_e')$). High end: the **transistor's internal junction capacitances** (Miller-multiplied C_{bc}) plus stray capacitance.

Check 1. (a) Power $\propto V^2$. At 0.707 of the voltage, the power is $0.707^2 = 0.5$ of maximum, and $20 \log 0.707 = -3$ dB. (b) **Down.** R_E returns to the AC path, so $A_v \approx -R_C/(r_e + R_E) = -560/565 \approx -1$. The overall gain is about 1 or less (this is Exp 8's effect). (c) No collector current: the transistor is cut off. Look for an open base connection, a missing emitter return, a wrong pin order (check the BC547's pinout: C-B-E with the flat face towards you) or a dead transistor.

Exp 7 · Emitter follower (common collector): verify its gain

Manual pp. 22–23 · hands-on (with LTspice steps in the manual's procedure) · CO4, CO5

5 BJT curves → 6 CE amplifier → **7 Emitter follower** → 8 Feedback

What you already have: from Exp 6, $r_e = 26\ mV/I_E$, and voltage-divider bias. From Exp 5, $V_E = V_B - 0.7\ V$ in the active region.

Two terms first:

- **Common collector (CC):** the collector sits at the supply, which is AC ground. Input at the base, output at the **emitter**.
- **Buffer:** a stage with gain ≈ 1, high input resistance and low output resistance. It copies a voltage without loading the source.

MANUAL EXP 7, AIM

Verify the gain of an emitter follower circuit. Viva form: "The gain is less than 1. Why would anyone build an amplifier that doesn't amplify voltage?"

GUESS FIRST

Hint: think about **current** and about **resistance seen from each side**.

How it works

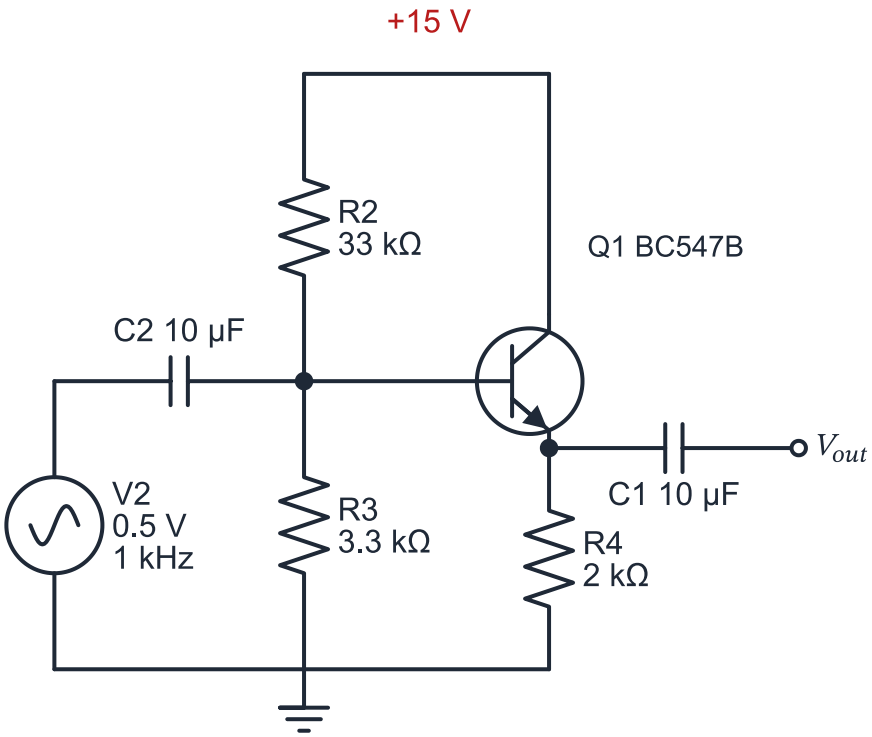


Fig 7.1, redrawn: 33 kΩ / 3.3 kΩ divider, 2 kΩ emitter resistor, output taken from the emitter through 10 µF.

The emitter is always one diode-drop below the base: $V_E = V_B - 0.7$. So when the base moves up by 0.5 V, the emitter moves up by (almost) 0.5 V. The emitter **follows** the base. That is the whole idea.

Q-point (LTspice .op): V_B = 1.36 V, V_E = 0.75 V, I_E = 0.75/2k = **0.376 mA**.

$$r_e = \frac{26\ \text{mV}}{0.376\ \text{mA}} = 69\ \Omega \qquad A_v = \frac{R_E}{R_E + r_e} = \frac{2000}{2069} = 0.967$$

Property	Value here	Why
voltage gain	≈ 0.97, **no inversion**	the emitter follows the base; the output is in phase
input resistance	≈ 3 kΩ (bias resistors), ≈ 600 kΩ into the base alone	the base sees R_E multiplied by β: $\beta(r_e + R_E)$ = 290 × 2.07 kΩ
output resistance	≈ 70 Ω (≈ r_e)	looking back into the emitter
current gain	large (≈ β)	the base current is tiny; the emitter current is β + 1 times bigger

So the answer to the viva question: it is a **current amplifier and impedance converter**. A weak, high-resistance source can drive a heavy, low-resistance load through it without its voltage sagging. That is why the extra design problem uses it as the output stage.

TRAP: THE MANUAL'S THEORY PARAGRAPH

It says the input impedance is "of the order of hundred ohms" and the output "of the order of ten ohms". The output part is right in spirit (tens of ohms). The input impedance is **high**: kilohms to hundreds of kilohms. High input, low output is the whole point.

DO IT

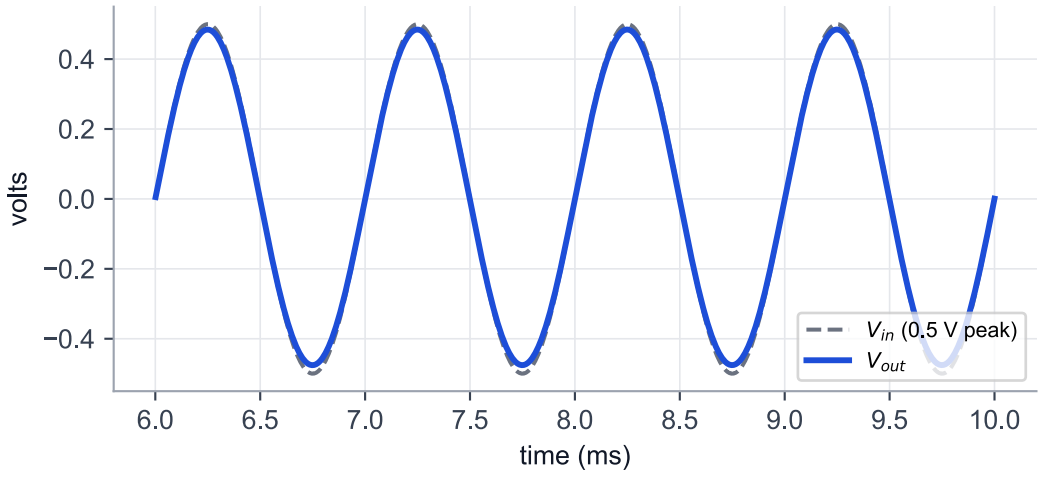
Hands-on:

1. Build Fig 7.1. Use the **15 V** shown in the figure. The parts list says 10 V; see the trap below.
2. Check the Q-point with no signal: $V_B \approx 1.36$ V, $V_E \approx 0.75$ V.
3. Generator: sine, 0.5 V peak, 1 kHz, into C2. DSO CH1 on the input, CH2 on V_{out}. The two traces should overlap, in phase.
4. Step the frequency through 200 Hz, 500, 1k, 10k, 100k, 1 MHz and record V_o. Gain = V_o/V_i.

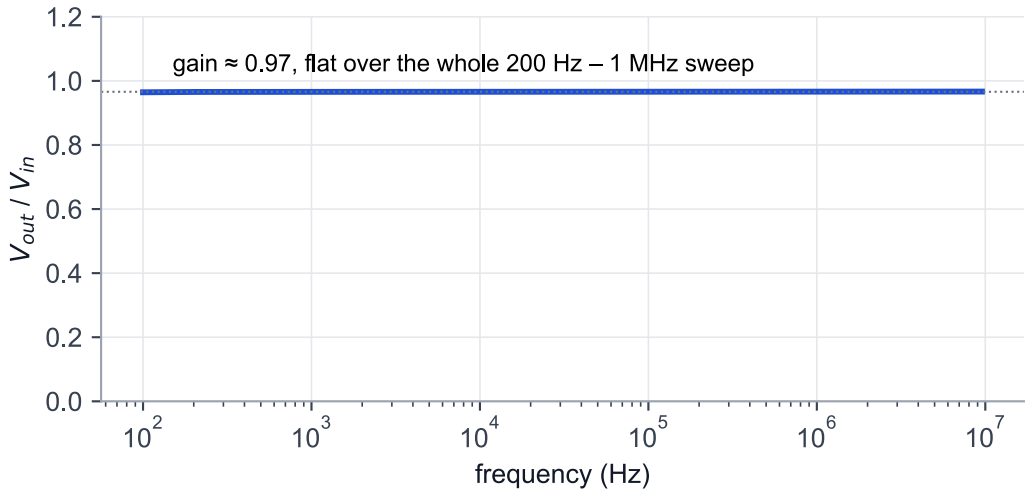
LTspice (manual's procedure):

1. V2 with `SINE(0 0.5 1k)` and AC amplitude `0.5`. Q1 = BC547B.
2. Run `.tran 10m` and compare V(out) with the input.
3. Then run `.ac dec 10 200 1Meg`. Comment out the line you are not using (right-click the directive and choose **Comment**), or the `;` prefix in the manual's figure does the same thing.

What you should see



LTspice: output (solid) sits on the input (dashed), in phase, 0.96 V p-p for 1.0 V p-p in.



Gain ≈ 0.97 across the whole 200 Hz – 1 MHz sweep. Neither −3 dB point is inside the sweep: f_L ≈ 5 Hz, f_H far above 1 MHz.

f (Hz)	200	500	1k	10k	100k	1M
A_v	0.965	0.966	0.966	0.966	0.966	0.966
dB	−0.31	−0.30	−0.30	−0.30	−0.30	−0.30

TRAP: A 10 V SUPPLY CLIPS THE OUTPUT

With 10 V, V_E drops to about 0.35 V. A 0.5 V-peak input then pulls the emitter below 0 V and the transistor cuts off: LTspice shows the negative half flattened at −0.32 V. With 10 V, either use 0.2 V peak (clean) or the figure's 15 V.

WRITE IN YOUR RECORD

Result: maximum gain of the emitter follower ≈ **0.97** (−0.3 dB), output in phase with input; flat from 200 Hz to 1 MHz, so the bandwidth exceeds the measured range (simulation: f_L ≈ 5 Hz).

Theory for the record: in the common-collector (emitter follower) circuit, the input is applied to the base and the output taken from the emitter, while the collector is at AC ground. Because $V_E = V_B - V_{BE}$, the emitter follows the base: voltage gain is slightly below 1 ($R_E/(R_E + r_e)$) with no phase inversion. The input resistance is high (≈ βR_E) and the output resistance low (≈ r_e), so the stage is used as a buffer for impedance matching from a high-impedance source to a low-impedance load. It has a high current gain and power gain, and a wide bandwidth.

CHECK 1

(a) Why is A_v always a little less than 1, never exactly 1? (b) Change R_E to 1 kΩ (same bias). Estimate the new I_E, r_e and A_v. (c) Which stage, CE or CC, would you put directly in front of an 8 Ω speaker, and why?

Answers

Opening question. It amplifies **current** (≈ β) and therefore power. It also converts a high source resistance into a low output resistance (a buffer), so a weak source can drive a heavy load.

Check 1. (a) The output is taken across R_E, which is in series with r_e: $A_v = R_E/(R_E + r_e)$. r_e is never zero. (b) I_E = 0.75/1k = 0.75 mA, r_e = 26/0.75 = 35 Ω, A_v = 1000/1035 = **0.966**: still about the same. (c) **CC (emitter follower).** Its output resistance is tiny, so the 8 Ω load does not collapse the voltage. A CE stage's output resistance (≈ R_C) would drop most of the signal internally.

Exp 8 · Effect of negative feedback (removing the emitter bypass capacitor)

Manual pp. 24–25 · LTspice · CO4, CO5 · the manual's two plots (≈ 42 dB and ≈ 4 dB) match the simulation below

5 BJT curves → 6 CE amplifier → 7 Emitter follower → **8 Feedback**

What you already have: from Exp 6, C_E bypasses R_E, and gain ≈ R_C/r_e. From Exp 7, a resistor in the emitter makes the output "follow" and pulls the gain down.

Two terms first:

- **Feedback:** feeding part of the output back to the input.
- **Negative** feedback opposes the input. It lowers the gain but steadies everything else.
- **Positive** feedback adds to the input. It leads to oscillation (Exp 11).
- **Current-series feedback:** the feedback voltage is produced by the output **current** ($I_E \approx I_C$ through R_E) and subtracted in **series** with the input ($V_{BE} = v_{in} - I_E R_E$).

MANUAL EXP 8, AIM

To observe the effect of negative feedback (emitter bypass capacitor) on the frequency response of an amplifier. Viva form: "Removing one capacitor cuts the gain from 122 to 1.7. Why would anyone want that?"

GUESS FIRST

Name one thing that gets **better** when the gain drops.

The idea

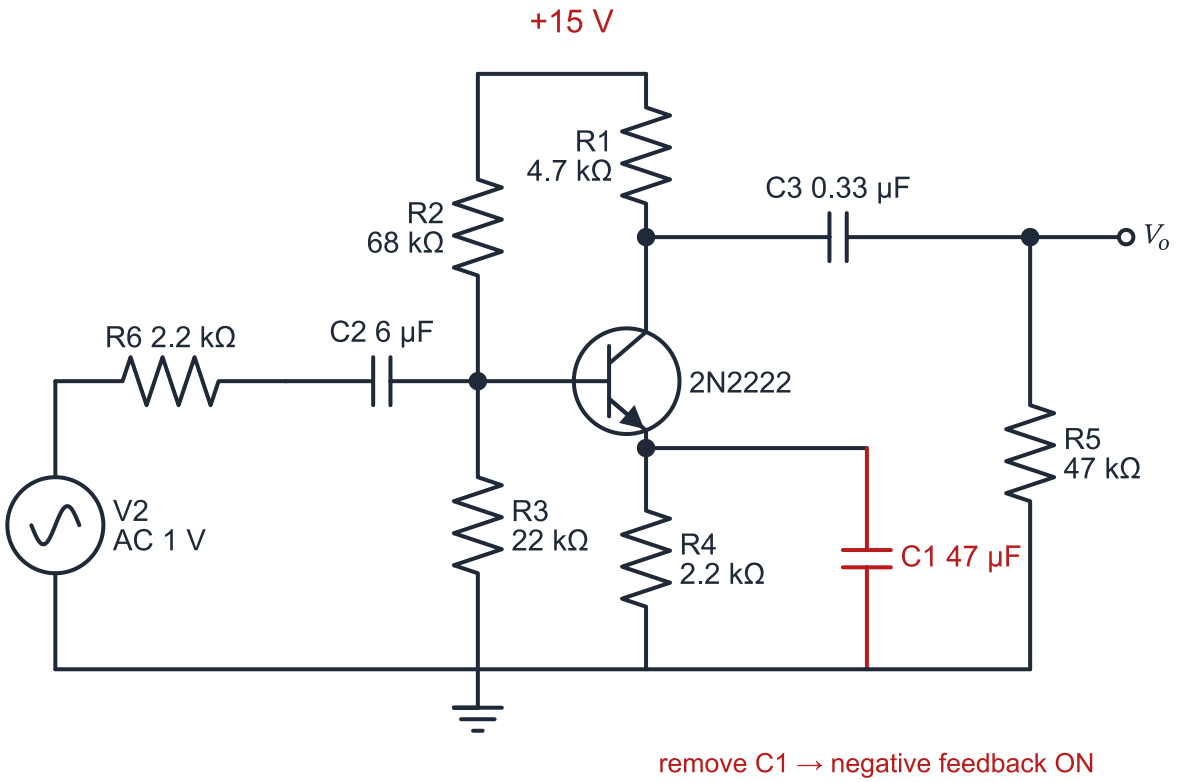


Fig 8.1, redrawn: a CE stage (2N2222) with load. C1 (red) is the bypass capacitor. Removing it is the whole experiment.

- **C1 in:** R4 is shorted for AC, so there is no feedback. $A_v \approx (R_C \parallel R_L)/r_e$, which is large.
- **C1 out:** the signal current through R4 (2.2 kΩ) makes a voltage that rises with the input, so the BE voltage the transistor actually sees shrinks. That is negative feedback. $A_v \approx (R_C \parallel R_L)/(r_e + R_E)$ = 4.27k/2.2k ≈ 1.9 at the base.

What negative feedback buys, at the cost of gain:

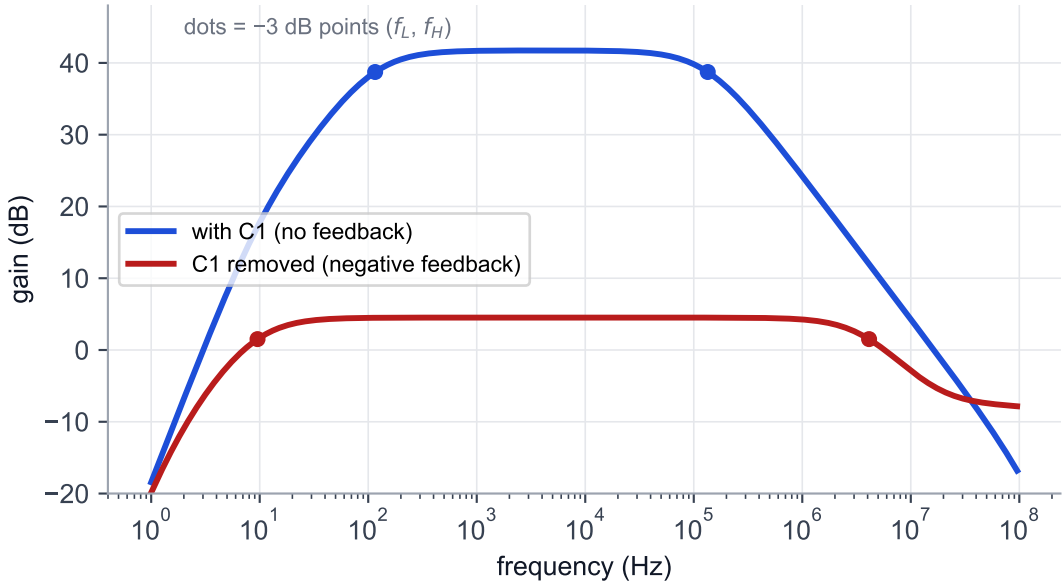
Property	Effect
gain	falls, but becomes **set by resistors** (R_C/R_E), not by β or temperature, so it is **stable**
bandwidth	**increases** (gain is traded for bandwidth)
distortion and noise	decrease
input resistance	increases (series feedback)

Do it

DO IT (LTSPICE)

1. Draw Fig 8.1. Q1 = **2N2222**. V2: AC amplitude `1`. Supply V1 = 15 V. Label the output `vo` (across R5, 47 kΩ).
2. Directive: `.ac dec 10 1 1Meg` (manual: 1 Hz – 1 MHz). Run. Plot V(vo): the solid line is the dB gain.
3. Read the peak dB with a cursor. Move the cursor to the frequencies where the gain is 3 dB lower: those are f_L and f_H.
4. **Delete C1** (or set it to `1p`). Run again and repeat.
5. Fill the table at 100 Hz, 500, 1k, 5k, 10k, 50k, 100k, 500k, 1M for both cases.

What you should see



LTspice. Blue: with C1, 41.7 dB peak, f_L 116 Hz, f_H 135 kHz. Red: without C1, 4.5 dB, f_L 9.5 Hz, f_H 4.1 MHz.

f (Hz)	100	500	1k	5k	10k	50k	100k	500k	1M
A_v with C1	79.8	119.1	121.4	122.1	121.9	114.5	98.0	31.7	16.3
A_{vf} without C1	1.67	1.68	1.68	1.68	1.68	1.68	1.68	1.67	1.63

	Without feedback (C1 in)	With feedback (C1 out)
maximum gain	122 (41.7 dB)	1.68 (4.5 dB)
f_L	116 Hz	9.5 Hz
f_H	135 kHz	4.1 MHz
bandwidth	≈ **135 kHz**	≈ **4.1 MHz**

Hand check (Q-point: V_B = 3.56 V, V_E = 2.90 V, so I_E = 1.32 mA and r_e = 19.7 Ω):

- R_{in} with feedback = 68k ∥ 22k ∥ β × 2.2k ≈ 16 kΩ. The source divider is 16/(16 + 2.2) = 0.88.
- A_{vf} ≈ 0.88 × 4.27k/2.22k = **1.70** (LTspice: 1.68) ✓.

WRITE IN YOUR RECORD

Result: bandwidth ≈ **135 kHz** without feedback (gain 41.7 dB). With negative feedback (C_E removed), the gain falls to 4.5 dB and the bandwidth rises to ≈ **4.1 MHz**: the higher cut-off is far above 1 MHz, as the manual expects.

Theory for the record: removing the emitter bypass capacitor leaves R_E unbypassed. The AC emitter current develops a voltage across R_E that subtracts from the input in series: current-series negative feedback. The voltage gain falls from ≈ R_C/r_e to ≈ $R_C/(r_e + R_E)$, but becomes stable (independent of β and temperature). The bandwidth increases, distortion and noise fall, and the input resistance rises. Negative feedback trades gain for bandwidth and stability.

CHECK 1

(a) Gain × bandwidth without feedback, and with feedback. Is the product the same? (b) Why is the 2N2222's own β almost irrelevant to the gain once C1 is out? (c) Which part of the response barely changed between the two cases at 100 kHz–1 MHz, and why?

Answers

Opening question. Negative feedback makes the gain **stable and predictable** (set by resistors, not β), **widens the bandwidth** (135 kHz → 4.1 MHz), and reduces distortion and noise. You add gain back with more stages; stability is hard to add later.

Check 1. (a) 122 × 135 kHz ≈ 16.5 MHz; 1.68 × 4.1 MHz ≈ 6.9 MHz. **Not the same**: "gain × bandwidth = constant" holds for a single dominant pole, not exactly for this circuit, but the direction (gain down, bandwidth up) holds. (b) $A_{vf} \approx (R_C \parallel R_L)/(r_e + R_E)$, and $R_E \gg r_e$, so β drops out. (c) The without-C1 curve is flat across it, while the with-C1 curve is already falling. With feedback, f_H moved out to MHz.

Exp 9 · JFET drain and transfer characteristics, and device parameters

Manual pp. 26–28 · LTspice · CO1, CO2 · theory: ECE2101 unit 3 · the simulation reproduces the manual's printed table exactly · the professor flagged these four parameters as an 8-mark question (classmate's report, 2026-10-05)

9 JFET curves → 10 CS amplifier → 11 Oscillator → 12 MOSFET inverter

What you already have: from Exp 1, a reverse-biased pn junction carries no current and has a depletion region with no free carriers. From Exp 5, families of output curves and slopes taken as parameters.

Three terms first:

- **FET (field-effect transistor):** current flows through a **channel**, and a **voltage** on the gate controls the channel's width. It is *voltage-controlled*. A BJT, by contrast, is controlled by base current.
- **JFET terminals: drain** (D, current in), **source** (S, current out), **gate** (G, controls).
- **Unipolar:** only one carrier type carries the current (electrons, in an n-channel JFET). A BJT uses both electrons and holes, which is why it is called *bipolar*.

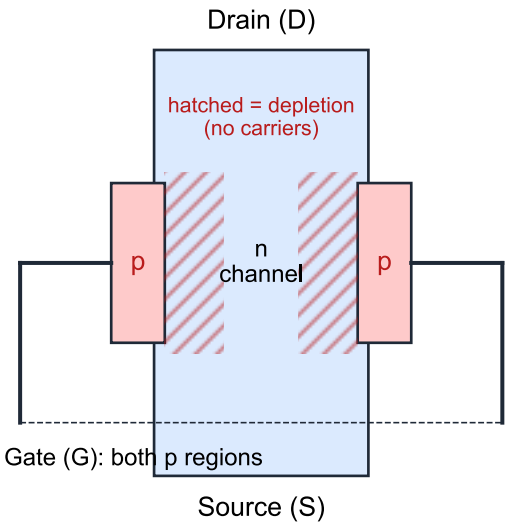
MANUAL EXP 9, AIM · PROFESSOR'S SLIDE

To plot the drain and transfer characteristics of a JFET in CS configuration and determine the device parameters. Viva / 8-mark form: "Define g_m, r_d, g_d and μ, find each from the curves, and show $\mu = g_m r_d$."

GUESS FIRST

Which of the four do you think is read from the *transfer* curve, and which from the *drain* curves?

How a JFET works



n-channel JFET: an n bar between two p gate regions. The gate–channel junctions are kept reverse-biased ($V_{GS} \le 0$).

1. **V_{GS} = 0:** the channel is fully open, and current flows from D to S.
2. **V_{GS} more negative:** the reverse-biased gate junctions' depletion regions widen into the channel, so the channel narrows and I_D falls.
3. **V_{GS} = V_P** (pinch-off voltage, ≈ −3 V here): the depletion regions meet and $I_D \approx 0$. The JFET is off.
4. **Raising V_{DS}** at fixed V_{GS}: at first I_D rises like a resistor (**ohmic region**). Then the channel pinches off near the drain and I_D levels out (**saturation**, or pinch-off region). The amplifier operates here.
5. The gate junction is reverse-biased, so **gate current ≈ 0**: the input resistance is enormous (≫ MΩ). This is the JFET's big advantage.

Shockley's equation (the transfer curve, in saturation):

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

- I_{DSS} = drain current with the gate **S**horted to the **S**ource (V_{GS} = 0), in saturation.

The four parameters (the professor's slide)

Parameter	Definition	Held constant	Read from
transconductance g_m	$\Delta I_D/\Delta V_{GS}$	V_{DS}	two drain curves at one V_{DS} (or the transfer curve's slope)
drain resistance r_d	$\Delta V_{DS}/\Delta I_D$	V_{GS}	one drain curve, in saturation
drain conductance g_d	$\Delta I_D/\Delta V_{DS} = 1/r_d$	V_{GS}	same
amplification factor μ	$-\Delta V_{DS}/\Delta V_{GS}$	I_D	**$\mu = g_m \times r_d$**

Why $\mu = g_m r_d$: μ asks how much V_{DS} change balances a V_{GS} change at constant I_D. Write it as a chain of ratios: $(\Delta V_{DS}/\Delta I_D) \times (\Delta I_D/\Delta V_{GS}) = r_d \times g_m$. The minus sign records that a more negative V_{GS} must be balanced by a larger V_{DS}.

Do it

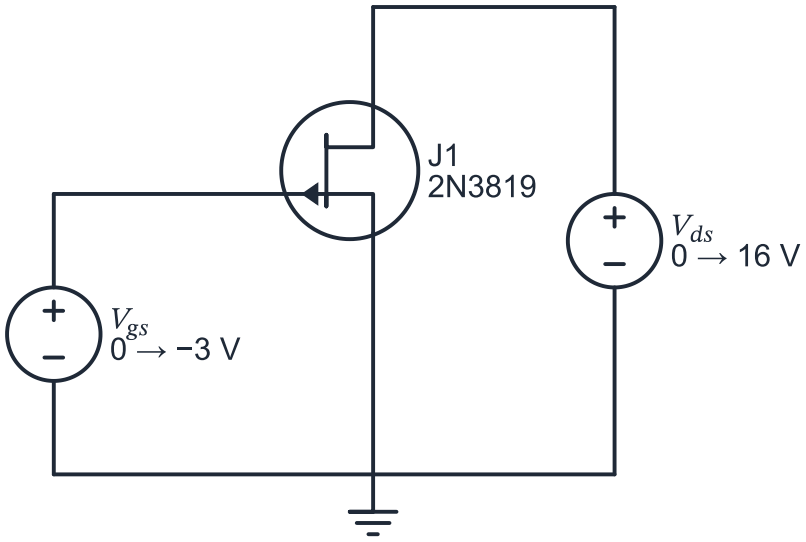
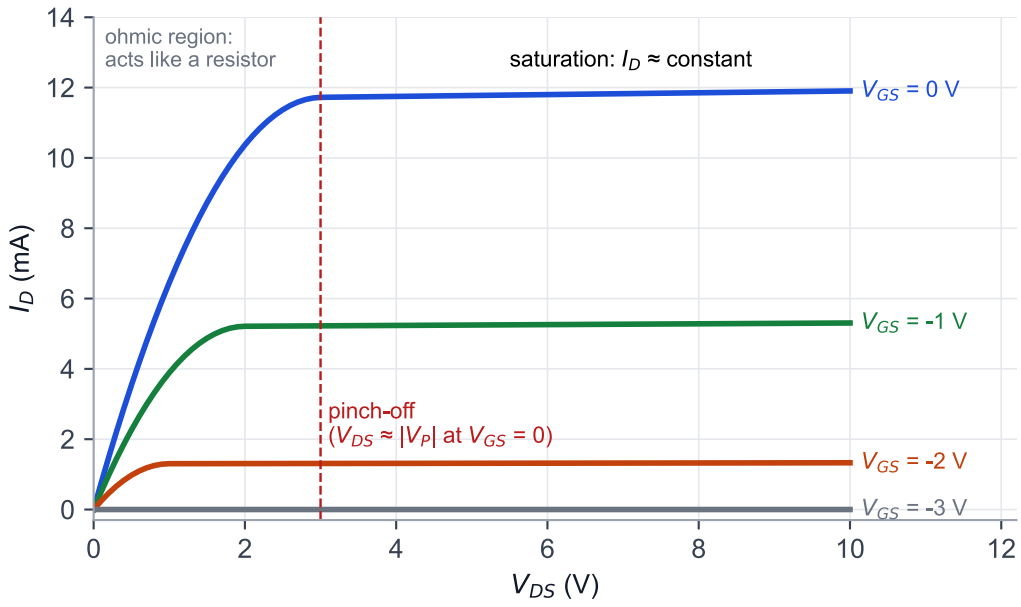


Fig 9.2, redrawn: V_{gs} on the gate, V_{ds} on the drain, source grounded.

DO IT (LTSPICE)

1. Draw the circuit with J1: Pick New JFET → **2N3819** (an n-channel JFET, `njf`). Name the sources `Vgs` and `Vds`.
2. **Drain characteristics:** `.dc Vds 0 16 0.1 Vgs 0 -3 -1`. This gives four curves, for V_{GS} = 0, −1, −2, −3 V. Plot **Id(J1)**.
3. **Transfer characteristic:** set Vds = 8 V, then `.dc Vgs -5 0 0.01`. Plot Id(J1) against V_{gs}.
4. Read I_{DSS} (cursor at V_{GS} = 0) and V_P (where I_D reaches ≈ 0).

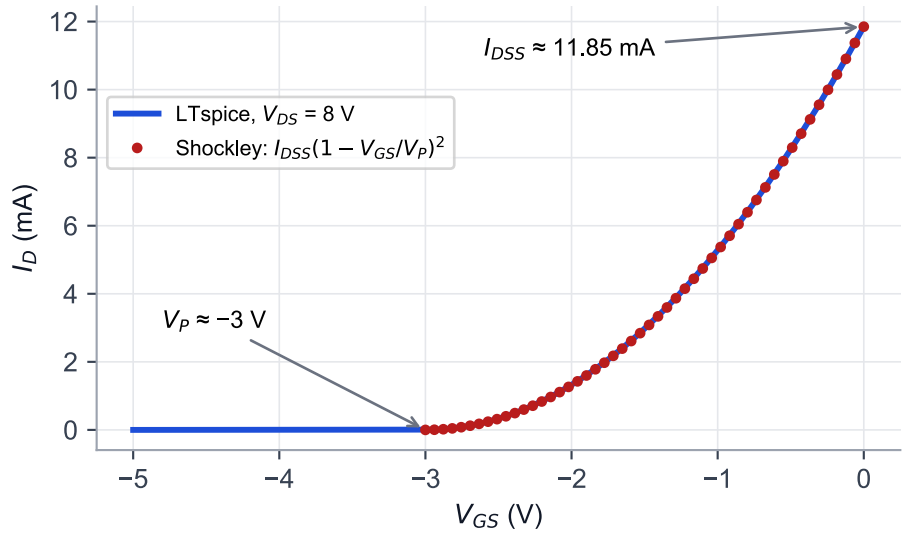
What you should see



Drain characteristics: ohmic rise, pinch-off, then flat saturation. Each curve flattens at $V_{DS} \approx V_{GS} - V_P$.

V_{DS} (V)	0	1	2	3	4	5	6	7	8
I_D (mA), V_{GS} = 0	0	6.45	10.37	11.72	**11.75**	11.77	**11.80**	**11.83**	11.85
I_D (mA), V_{GS} = −1	0	3.89	5.21	5.22	**5.24**	5.25	5.26	5.27	5.28
I_D (mA), V_{GS} = −2	0	1.30	1.31	1.31	1.31	1.32	1.32	1.32	1.32
I_D (mA), V_{GS} = −3	0	0	0	0	0	0	0	0	0

These are the manual's printed numbers (it lists them in amps: 0.01185 A = 11.85 mA).



Transfer characteristic at V_{DS} = 8 V, with Shockley's equation (dots) laid on top: a near-perfect match.

Calculations

- **I_{DSS} = 11.85 mA** (V_{GS} = 0, V_{DS} = 8 V).
- **V_P ≈ −3 V** (I_D reaches 0 at V_{GS} = −3 V).

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{11.75 - 5.24 \text{ mA}}{0 - (-1) \text{ V}} = 6.51 \text{ mA/V} (= 6.51 \text{ mS}) \qquad \text{at } V_{DS} = 4 \text{ V}$$

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{8 - 3 \text{ V}}{11.852 - 11.722 \text{ mA}} = \frac{5 \text{ V}}{0.130 \text{ mA}} = 38.5 \text{ k}\Omega \qquad \text{at } V_{GS} = 0$$

$$g_d = \frac{1}{r_d} = 26 \ \mu\text{S} \qquad \mu = g_m r_d = 6.51 \text{ mS} \times 38.5 \text{ k}\Omega \approx 250$$

Cross-check with Shockley:

- The slope at $V_{GS} = 0$ is $g_{m0} = 2I_{DSS}/|V_P| = 2 \times 11.85/3 = 7.9$ mS.
- The 0 → −1 V chord equals the slope at the midpoint, $V_{GS} = -0.5$ V: $g_m = g_{m0}(1 - 0.5/3) = 6.6$ mS ✓ (6.51 measured).

TRAP: THE MANUAL'S R_D LINE

It writes (7 − 6)/(0.1183 − 0.1180) = 3.3 kΩ. The currents are 0.01183 A and 0.01180 A, so the difference is 0.03 mA, and r_d = 1 V / 0.03 mA = **33 kΩ**. Also, a 1 V span with 4-digit data is coarse: use a wide V_{DS} span (3 → 8 V) inside saturation, which gives 38.5 kΩ.

WRITE IN YOUR RECORD

Result: I_{DSS} = **11.85 mA**, V_P = **−3 V**, g_m = **6.51 mS** (V_{DS} = 4 V), r_d = **38.5 kΩ** (V_{GS} = 0); also g_d = 26 µS and $\mu = g_m r_d \approx 250$.

Theory for the record: an n-channel JFET is an n-type channel between two p-type gate regions. With the gate–source junction reverse-biased, the depletion regions narrow the channel as V_{GS} becomes more negative, so the gate voltage controls the drain current. Gate current is negligible, giving very high input resistance. As V_{DS} increases, I_D first rises linearly (ohmic region), then the channel pinches off and I_D saturates. At $V_{GS} = V_P$ the channel is cut off. In saturation, $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ (Shockley). Parameters: $g_m = \Delta I_D/\Delta V_{GS}$ at constant V_{DS}; $r_d = \Delta V_{DS}/\Delta I_D$ at constant V_{GS}; $\mu = g_m r_d$.

CHECK 1

(a) Use Shockley to predict I_D at $V_{GS} = -2$ V, and compare with the table. (b) Why is a JFET's gate never forward-biased in normal use? (c) BJT vs JFET: give two differences an examiner would accept.

Answers

Opening question. g_m from the **transfer** curve (or two drain curves at one V_{DS}); r_d and g_d from **one drain curve** in saturation; μ from $g_m \times r_d$.

Check 1. (a) $11.85 \times (1 - 2/3)^2 = 11.85/9 =$ **1.32 mA**; the table gives 1.31–1.32 ✓. (b) Forward bias would make the gate junction conduct. The gate would draw current, the high input resistance would be lost, and the device could be damaged. (c) Any two: a JFET is voltage-controlled, a BJT current-controlled; a JFET is unipolar, a BJT bipolar; a JFET has very high input resistance, a BJT moderate ($h_{ie} \approx$ kΩ); a JFET is less noisy and more thermally stable; a BJT has higher gain (g_m).

Exp 10 · JFET common-source amplifier: frequency response

Manual pp. 29–30 · LTspice · CO4, CO5 · the JFET is the 2N5434 (the manual's "2N54349" is a typo)

9 JFET curves → **10 CS amplifier** → 11 Oscillator → 12 MOSFET inverter

What you already have: the BJT's CE amplifier (Exp 6), meaning bias, coupling and bypass capacitors, gain, and −3 dB points. The JFET's g_m (Exp 9).

Two terms first:

- **Common source (CS):** the FET's version of CE. Input at the gate, output at the drain, source common.
- **Self-bias:** the source resistor R_s lifts the source above the gate, which makes V_{GS} negative without a negative supply.

MANUAL EXP 10, AIM

To plot the frequency response of a common-source amplifier. Viva form: "Estimate the gain from g_m and the resistors, and say why the output is 180° out of phase."

GUESS FIRST

CE gain was R_C/r_e. What do you think replaces $1/r_e$ for a FET?

The circuit and its bias

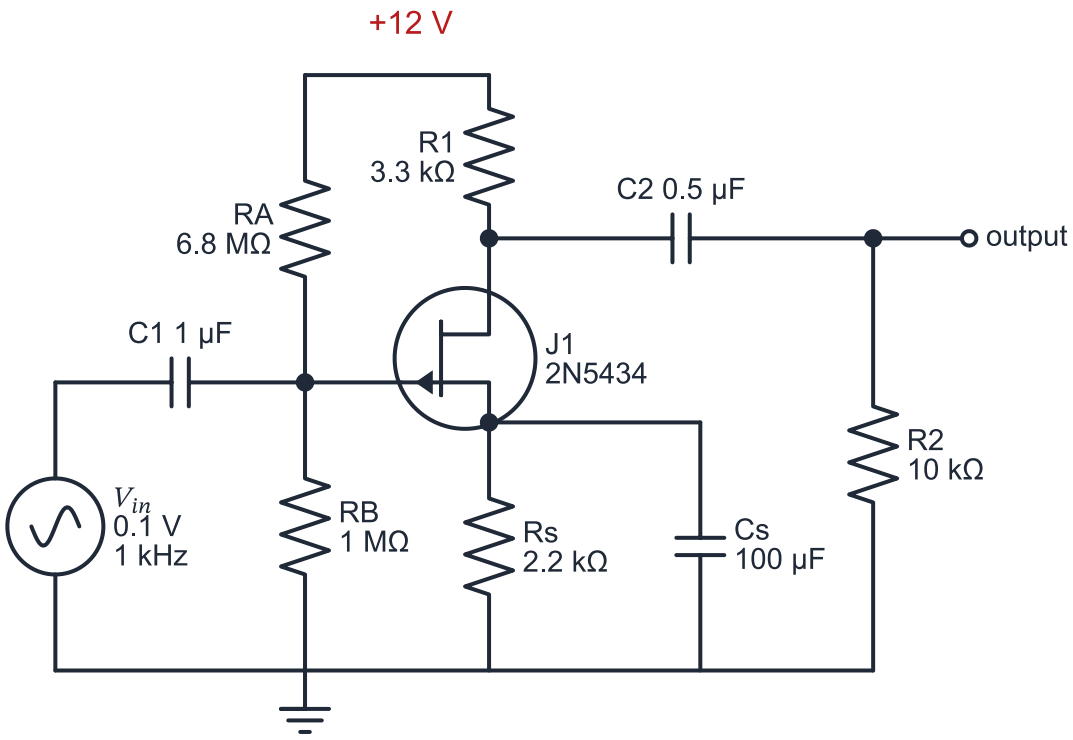


Fig 10.1, redrawn: gate divider RA/RB, drain resistor R1, source resistor Rs bypassed by Cs, 10 kΩ load through C2.

Part	Job
RA 6.8 MΩ, RB 1 MΩ	gate divider: $V_G = 12 \times 1/7.8 \approx 1.54$ V. Megohms are fine because the gate draws no current.
Rs 2.2 kΩ	self-bias: $V_{GS} = V_G - I_D R_s$ is negative
Cs 100 µF	bypasses Rs for AC, giving full gain (as C_E did in Exp 6)
R1 3.3 kΩ	drain load: converts the I_D change into a voltage change
C1, C2	coupling: block DC, pass the signal

Q-point (LTspice .op): $V_G = 1.54$ V, $V_S = 3.17$ V, so **$V_{GS} = -1.63$ V**. $I_D = 3.17/2.2\text{k} =$ **1.44 mA**, and $V_D = 7.25$ V. The drain sits well above the source, so the JFET is in saturation ✓.

Gain by hand: the small-signal model is a current source $g_m v_{gs}$ into the drain.

$$A_v = -g_m (R_1 \parallel R_2) = -g_m \times (3.3\ \text{k}\Omega \parallel 10\ \text{k}\Omega) = -g_m \times 2.48\ \text{k}\Omega$$

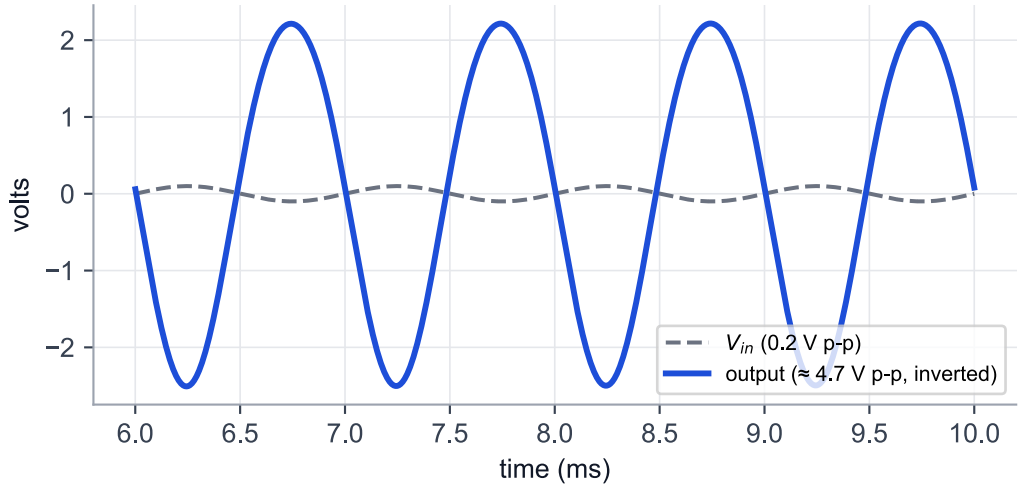
For the 2N5434 at this Q-point, $g_m \approx 9.7$ mS (from its model: $I_{DSS} \approx 65$ mA, $V_P \approx -1.9$ V, using $g_m = (2I_{DSS}/|V_P|)(1 - V_{GS}/V_P)$). So $A_v \approx -24$. The **minus sign** is the 180° phase inversion: as v_{gs} goes up, I_D goes up, and the drain voltage $V_{DD} - I_D R_1$ goes **down**.

Do it

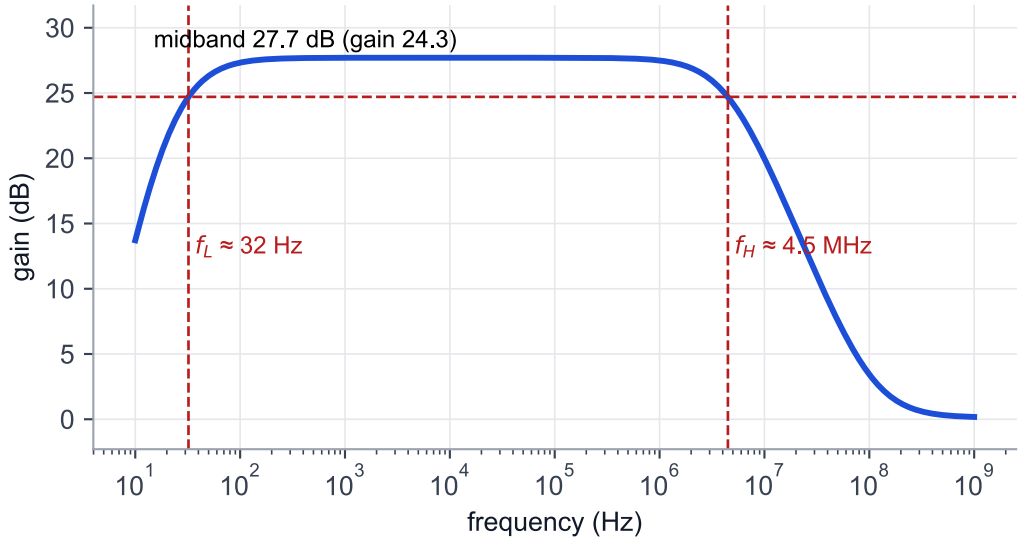
DO IT (LTSPICE)

1. Draw Fig 10.1 with J1 = **2N5434**, VDD = 12 V. Name the output node `out` (across R2).
2. Vin: `SINE(0 0.1 1k)` with AC amplitude `1`.
3. **Transient:** `.tran 10m`. Plot the input and V(out), and read both peak-to-peak values with cursors.
4. **Frequency response:** `.ac dec 10 100 100Meg`. Read the midband dB and find the two −3 dB frequencies.
5. Fill the table at 100 Hz, 1k, 5k, 10k, 50k, 100k, 500k, 1M, 5M, 10M, 100M.

What you should see



Transient: 0.2 V p-p in, ≈ 4.7 V p-p out, inverted.



AC: midband gain 24.3 (27.7 dB), f_L ≈ 32 Hz, f_H ≈ 4.5 MHz.

f (Hz)	100	1k	5k	10k	50k	100k	500k	1M	5M	10M	100M
gain	23.3	24.3	24.3	24.3	24.3	24.3	24.1	23.7	16.2	10.0	1.5
dB	27.3	27.7	27.7	27.7	27.7	27.7	27.6	27.5	24.2	20.0	3.4

Calculation

- Transient gain = $V_{O(pp)}/V_{in(pp)}$ = 4.73/0.2 = **23.6**.
- AC midband gain = **24.3**.
- BW = $f_H - f_L$ = 4.48 MHz − 32 Hz ≈ **4.5 MHz**.

The low cut-off comes from Cs (with $R_s \parallel 1/g_m \approx 100\ \Omega$) and from C2 (with R1 + R2). The high cut-off comes from the JFET's gate–drain and gate–source capacitances (35 pF each in its model), Miller-multiplied.

WRITE IN YOUR RECORD

Result: maximum gain ≈ **24.3 (27.7 dB)**; transient gain 4.73 V/0.2 V = 23.6; output 180° out of phase; f_L ≈ 32 Hz, f_H ≈ 4.5 MHz, bandwidth ≈ **4.5 MHz**.

Theory for the record: in a common-source amplifier the signal is applied to the gate and taken from the drain, with the source common (bypassed to AC ground). The gate divider and source resistor set a negative V_{GS} that biases the JFET in saturation. A gate-voltage change produces a drain-current change $g_m v_{gs}$, which develops an output voltage across the drain load: $A_v = -g_m(R_D \parallel R_L)$. The output is inverted (180°). The input resistance is very high (set by the gate divider). The gain is lower than a BJT CE stage, but the input resistance and noise performance are better. Coupling/bypass capacitors set f_L; device capacitances set f_H.

CHECK 1

(a) Remove Cs. What happens to the gain, and why? (b) Why can the gate resistors be megohms here, when the BJT's bias resistors were kilohms? (c) Estimate the input resistance seen by the source.

Answers

Opening question. g_m replaces $1/r_e$: $A_v = -g_m(R_D \parallel R_L) \approx -24$. The output is inverted because a rising gate voltage raises I_D, which increases the drop across R1 and lowers the drain voltage.

Check 1. (a) The gain drops: R_s now gives negative feedback (as in Exp 8). $A_v \approx -g_m R_D'/(1 + g_m R_s) = -24/(1 + 9.7\text{m} \times 2.2\text{k}) \approx$ **−1.1**. (b) The JFET gate takes no current, so the divider is not loaded. A BJT's base current would make megohm resistors drop volts and wreck the bias. (c) $R_{in} \approx RA \parallel RB = 6.8\ \text{M}\Omega \parallel 1\ \text{M}\Omega \approx$ **0.87 MΩ**.

Exp 11 · RC phase-shift oscillator

Manual pp. 31–33 · LTspice · CO4 · the manual's own observed frequency (13.76 kHz) and the simulation below (13.96 kHz) agree within 1.5 %

9 JFET curves → 10 CS amplifier → 11 Oscillator → 12 MOSFET inverter

What you already have:

- From Exp 6, a CE stage inverts its signal (180°) and has gain.
- From Exp 8, negative feedback opposes the input.
- From C&S, in an RC circuit the voltage across R leads the input by an angle that depends on frequency.

Two terms first:

- **Oscillator:** a circuit that produces a steady AC output with **no** AC input. It is an amplifier whose output is fed back to its own input **in phase** (positive feedback).
- **Barkhausen criterion:** for steady oscillation, the loop gain must be $|A\beta| = 1$ (slightly more than 1 to start up), and the total phase shift around the loop must be **360°** (or 0°).

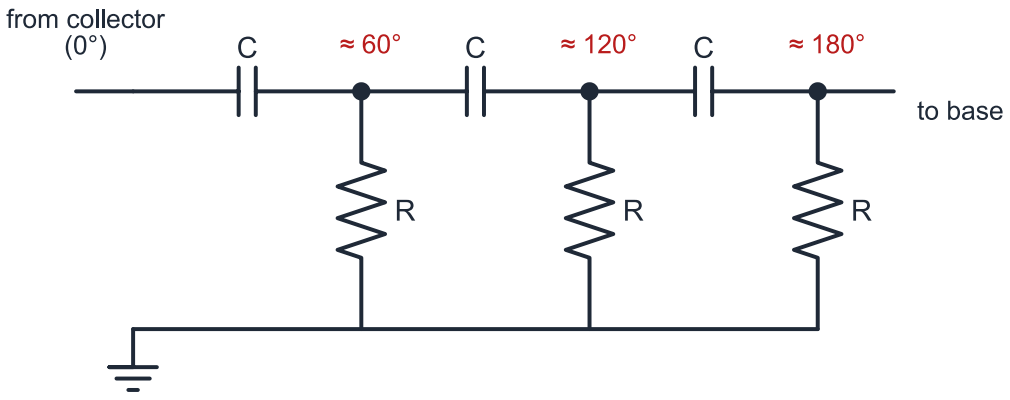
MANUAL EXP 11, AIM

To observe the frequency of oscillation of a phase-shift oscillator. Viva form: "Why three RC sections, and what is the frequency for R = 10 kΩ, C = 0.5 nF?"

GUESS FIRST

The transistor gives 180°. How much must the RC network give, and how many degrees can one RC section give at most?

How it works



The feedback network: three CR sections, each adding phase lead. At one frequency the total is exactly 180°.

- The CE transistor inverts: **180°**. The RC network must add the other **180°** at the oscillation frequency, so the loop totals 360°.
- One RC section can give at most 90°, and only at f → 0, where the signal is gone. So **three** sections are used, about 60° each (in the manual's simple picture), to make 180° at a usable frequency.
- At exactly that frequency, the ladder's output is 1/29 of its input. So the **amplifier gain must be ≥ 29** for the loop gain to reach 1.

$$f = \frac{1}{2\pi RC\sqrt{(2N)}} = \frac{1}{2\pi RC\sqrt{(6)}} \qquad (N = 3 \text{ sections})$$

For R = 10 kΩ, C = 0.5 nF:

$$f = \frac{1}{2\pi \times 10^{4} \times 0.5 \times 10^{-9} \times 2.449} = 13.0 \text{ kHz}$$

For C = 1 nF: **6.50 kHz**.

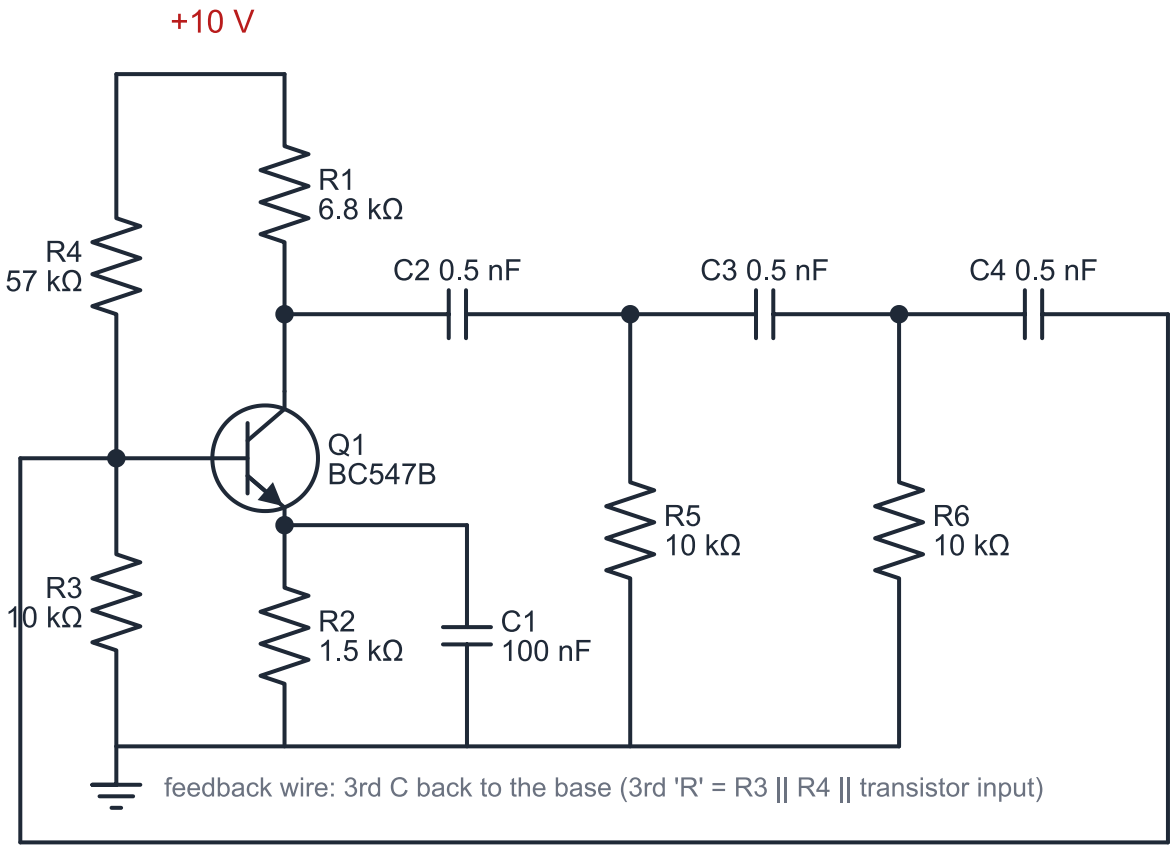
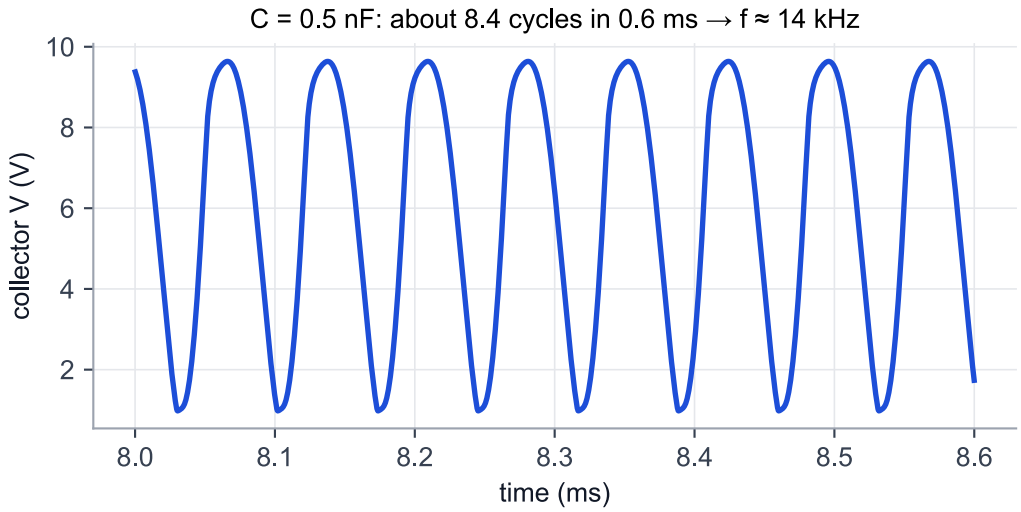


Fig 11.1, redrawn: a BC547B CE stage (10 V) with a three-section CR ladder from collector back to base.

DO IT (LTSPICE)

1. Draw Fig 11.1: Q1 = BC547B; R4 57k, R3 10k (bias); R1 6.8k (collector); R2 1.5k with C1 100n (emitter); ladder C2/C3/C4 = 0.5n, R5/R6 = 10k; V1 = 10 V. The third C goes back to the **base**.
2. Directive: `.tran 0 5m 2.5m` (simulate 5 ms, and plot only from 2.5 ms, after start-up). The circuit starts oscillating on its own. If the trace stays flat, add `startup` at the end of the directive to kick it.
3. Plot the collector voltage. Place cursor 1 on one peak and cursor 2 on a peak N cycles later. The readout gives the time difference and **Freq** (1/Δt). Divide the time difference by N for one period.
4. Change all three capacitors to `1n` and repeat.

What you should see



LTspice collector voltage (C = 0.5 nF): a steady oscillation of about 8.7 V p-p.

S.No.	R	C	Observed f (LTspice)	Theoretical f	% error
1	10 kΩ	0.5 nF	13.96 kHz	13.00 kHz	7.4 %
2	10 kΩ	1 nF	8.40 kHz	6.50 kHz	29 %

$$\% \text{ error} = \frac{|f_{theory} - f_{observed}|}{f_{theory}} \times 100 = \frac{|13.00 - 13.96|}{13.00} \times 100 = 7.4\,\%$$

Why the error exists (and grows): the formula assumes three **identical** sections and an ideal amplifier that doesn't load them. In the real circuit:

- the first C is driven through the 6.8 kΩ collector resistor, not from a zero-resistance source;
- the "third R" is not 10 kΩ but the base side: R3 ∥ R4 ∥ the transistor's input resistance, about 1–2 kΩ.

So the real network differs from the ideal one, and so does its 180° frequency. A % error of several percent is normal. LTspice shows the mismatch is worse at 1 nF.

WRITE IN YOUR RECORD

Result: the phase-shift oscillator produced **13.96 kHz** (C = 0.5 nF; theory 13.0 kHz, error 7.4 %) and **8.40 kHz** (C = 1 nF; theory 6.50 kHz). Increasing C (or R) lowers the frequency, as f ∝ 1/RC predicts.

Theory for the record: an oscillator is an amplifier with positive feedback that sustains an output without any input. The Barkhausen criterion requires a loop gain of at least 1 and a total loop phase shift of 360°. In the RC phase-shift oscillator, a CE transistor stage provides 180°, and a three-section RC ladder provides another 180° at $f = 1/(2\pi RC\sqrt{6})$. The ladder attenuates by 1/29 at that frequency, so the amplifier gain must be at least 29. Using more sections (e.g. four, 45° each) improves frequency stability. Varying R or C tunes the frequency. Used for audio-frequency sine generation.

CHECK 1

(a) Why can't two RC sections work? (b) The amplifier's gain is 20. What happens? (c) Calculate f for R = 4.7 kΩ, C = 1 nF.

Answers

Opening question. The network must add 180°. One section gives less than 90° (90° only at zero frequency), so at least three are needed: ≈ 60° each in the simple picture. $f = 1/(2\pi \times 10\text{k} \times 0.5\text{n} \times \sqrt{6}) =$ **13.0 kHz**.

Check 1. (a) Two sections reach 180° only at $f \to 0$, where the signal is fully attenuated, so the loop can never meet the phase condition at a usable frequency. (b) Loop gain = 20/29 < 1: any oscillation dies out and the output stays flat. (c) $f = 1/(2\pi \times 4700 \times 10^{-9} \times 2.449) =$ **13.8 kHz**.

Exp 12 · MOSFET inverter (resistive load)

Manual pp. 34–35 · LTspice · CO4 · theory: ECE2101 unit 4 (MOSFETs) · links straight to the Digital Electronics lab (logic gates)

9 JFET curves → 10 CS amplifier → 11 Oscillator → **12 MOSFET inverter**

What you already have:

- From Exp 9, a FET's gate voltage controls its channel.
- From Exp 5, a transistor can be a switch: cut-off = open, saturation = closed.
- From digital electronics: NOT gate, logic 1 and logic 0.

Three terms first:

- **Enhancement nMOS:** a MOSFET whose gate is insulated by oxide (so it draws **no** gate current). It has **no** channel at $V_{GS} = 0$. A channel forms only when $V_{GS} > \mathbf{V_{TH}}$ (threshold voltage).
- **Inverter:** a NOT gate. Input high gives output low, and input low gives output high.
- **VTC (voltage transfer characteristic):** V_{out} plotted against V_{in}.

MANUAL EXP 12, AIM

Analyse an inverter circuit using a MOSFET. Viva form: "Explain the two states of the circuit, and mark V_{OH}, V_{OL}, V_{IL}, V_{IH} and V_M on the VTC."

GUESS FIRST

When the input is 0 V, is the MOSFET on or off? What is the output then?

How it works

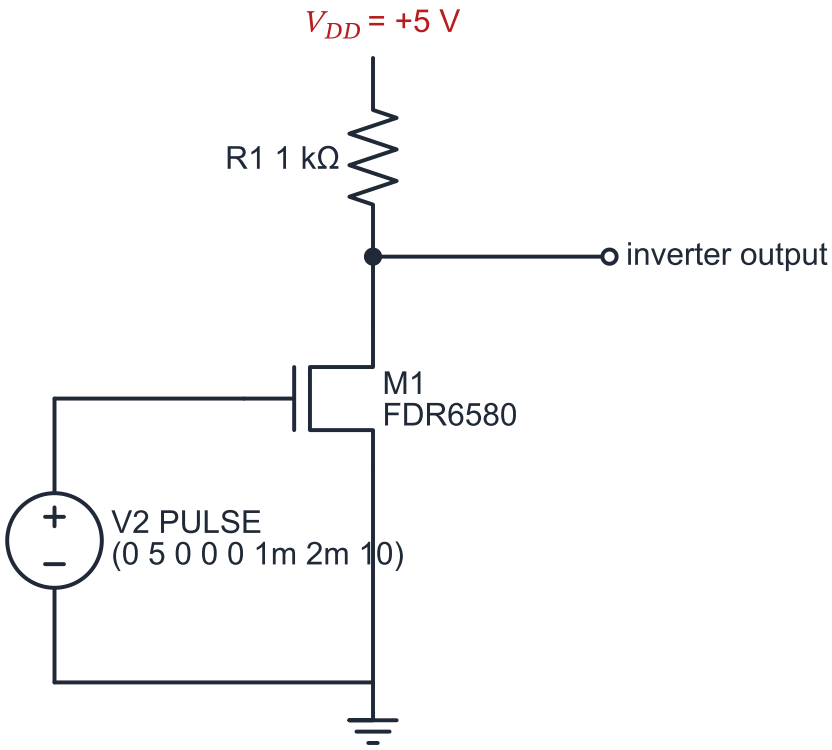


Fig 12.1, redrawn: 1 kΩ from +5 V to the drain (output); input pulse on the gate; source to ground.

Input	MOSFET	Current	Output
0 V (logic 0), below V_{TH}	**off** (cut-off): no channel	I_D = 0, so no drop across the 1 kΩ	$V_{out} = V_{DD}$ = **5 V (logic 1)**
5 V (logic 1), above V_{TH}	**on**, fully (deep triode): the channel is a very low resistance (≈ 9 mΩ for this part)	I_D ≈ 5 V/1 kΩ = 5 mA	V_{out} ≈ 5 mA × 9 mΩ ≈ **0 V (logic 0)**

VTC terms:

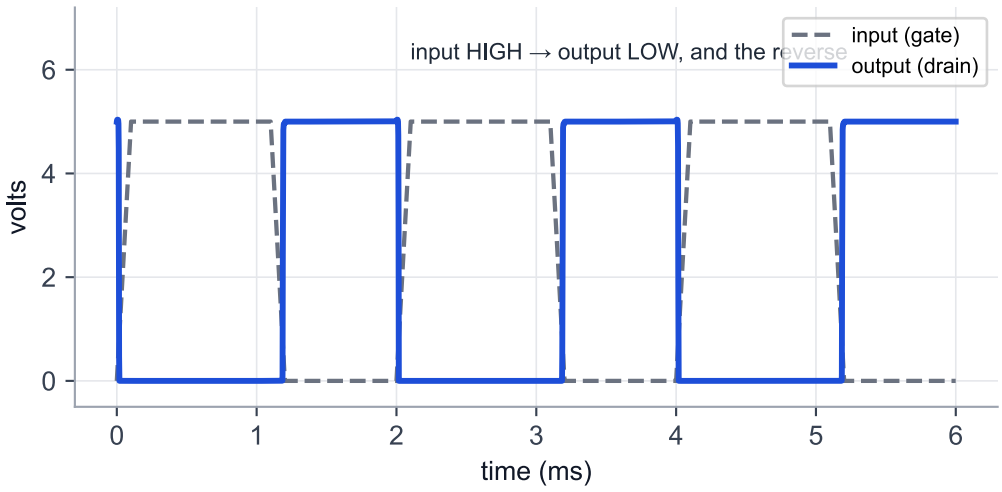
Term	Meaning
V_{OH} / V_{OL}	output high / output low levels
V_{IL}	the highest input still read as 0. Where the VTC slope first reaches −1.
V_{IH}	the lowest input read as 1. Where the slope returns to −1.
V_M	switching threshold, where $V_{out} = V_{in}$
noise margins	$NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$: how much noise an input can carry before the gate misreads it

Do it

DO IT (LTSPICE)

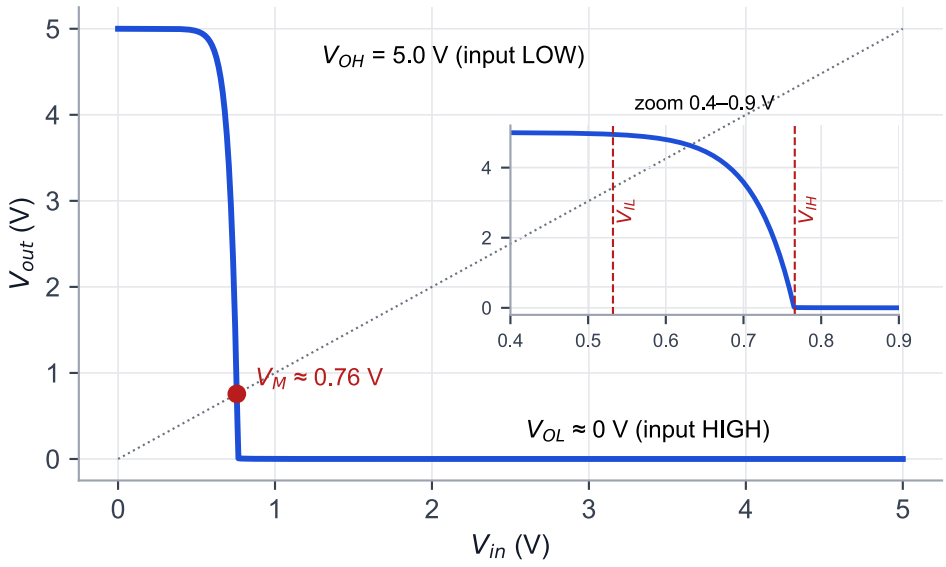
1. Draw Fig 12.1: M1: Pick New MOSFET → **FDR6580**; R1 = 1k; V1 = 5 V.
2. V2: **PULSE**: V1 = 0, V2 = 5, Tdelay = 0, Trise = 0, Tfall = 0, Ton = 1m, Tperiod = 2m, Ncycles = 10, which gives `PULSE(0 5 0 0 0 1m 2m 10)`.
3. Directive: `.tran 0 20m 0 1u`. The manual's bare `.tran 20m` works too, but its coarse steps show slanted edges and a false spike up to about 6.8 V.
4. Plot the input and output. Set the y-axis limits manually to −0.5 … 5.5 V. Read V_{min} and V_{max} of each.
5. **VTC (extra marks):** replace the pulse with a DC source, run `.dc V2 0 5 0.001`, and plot V(out) against V2.

What you should see



Transient: the output is the input upside-down: 5 V when the input is 0, about 0 V when the input is 5 V.

Voltage	V_{min}	V_{max}
V_{in}	0 V	5 V
V_{out}	≈ 0 V (0.04 mV)	5.0 V



VTC from LTspice: a very sharp switch near 0.76 V (zoom inset).

V_{OH}	V_{OL}	V_{IL}	V_{IH}	V_M	NM_L	NM_H
5.00 V	≈ 0 V	0.53 V	0.77 V	0.76 V	0.53 V	4.23 V

Why it switches so low and so sharply: the FDR6580 is a **power** MOSFET. It conducts amps from a tiny gate overdrive, so just past V_{TH} it already pulls the 5 mA that the 1 kΩ allows. The switch therefore happens right at V_{TH}, far below $V_{DD}/2$.

The manual's Fig 12.2 (with $V_{IL} \approx 0.9$ V and $V_{IH} \approx 2.2$ V) is a generic textbook VTC, not this part's.

TRAP: V_{TH} VS V_M

The manual says "V_{th} is the inverter threshold voltage, which is $V_{dd}/2$". These are two different quantities:

- **V_{TH}** is the MOSFET's own turn-on voltage.
- **V_M** is the inverter's switching point (where $V_{out} = V_{in}$).

$V_M = V_{DD}/2$ is the **design goal** of a symmetric CMOS inverter. In this resistor-load circuit, $V_M \approx 0.76$ V.

WRITE IN YOUR RECORD

Result: the output is inverted with respect to the input, with maximum voltage **5 V** as logic 1 and minimum voltage **≈ 0 V** as logic 0. (VTC: $V_{IL} = 0.53$ V, $V_{IH} = 0.77$ V, $V_M = 0.76$ V.)

Theory for the record: the inverter is the basic digital gate; NAND, NOR and larger logic are built from the same idea. In the resistive-load nMOS inverter, an enhancement nMOS pulls the output down and a resistor pulls it up to V_{DD}. For $V_{in} < V_{TH}$ the MOSFET is cut off, no current flows, and $V_{out} = V_{DD}$ (logic 1). For a high input the MOSFET conducts in the linear region and pulls the output close to 0 (logic 0). The VTC shows V_{OH}, V_{OL} and the transition region, with V_{IL} and V_{IH} at the unity-slope points and V_M where $V_{out} = V_{in}$. Noise margins are $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$. A drawback: current (and power) flows continuously when the output is low, which CMOS removes.

CHECK 1

(a) How much power does this inverter waste when its output is LOW? When HIGH? (b) Why does the MOSFET's gate draw no DC current? (c) What replaces the resistor in a CMOS inverter, and what does that fix?

Answers

Opening question. Input 0 V: the MOSFET is **off**, so no current flows and the output = **5 V**. Input 5 V: the MOSFET is **on** and the output ≈ 0 V. V_{OH} = 5 V, V_{OL} ≈ 0, V_{IL} = 0.53 V, V_{IH} = 0.77 V, V_M = 0.76 V.

Check 1. (a) LOW: 5 mA flows from the 5 V supply, so P = 5 V × 5 mA = **25 mW**, nearly all in the resistor. HIGH: no current, so ≈ 0 W. (b) The gate is insulated from the channel by a thin oxide layer: a capacitor, which passes no DC. (c) A **pMOS** transistor (pull-up). Only one transistor is on in each state, so no DC current flows from V_{DD} to ground. The static power becomes ≈ 0, and V_{OL} is exactly 0.

Extra · Two-stage RC-coupled amplifier, and the design problem

Manual pp. 3 and 36–38 · LTspice · the design answer is verified spec-by-spec in LTspice (src/sim/design.py)

6 CE amplifier → 7 Emitter follower → 8 Feedback → **Extra: cascade + design**

What you already have:

- From Exp 6, one CE stage: gain ≈ R_C/r_e, bandwidth f_L to f_H.
- From Exp 7, the emitter follower: low output resistance.
- From Exp 8, unbypassed R_E lowers gain and raises input resistance.

Two terms first:

- **Cascade:** stages in series, each one's output feeding the next one's input.
- **RC coupling:** the link between stages is a capacitor (it blocks one stage's DC from the next) plus the next stage's bias resistors.

Part 1 · Two-stage RC-coupled amplifier

MANUAL, ADDITIONAL EXPERIMENT

To measure the gain and plot the frequency response of a two-stage RC-coupled amplifier and determine its 3 dB bandwidth. Viva form: "Two stages: does the gain add or multiply, and what happens to the bandwidth?"

GUESS FIRST

Answer both halves before reading on.

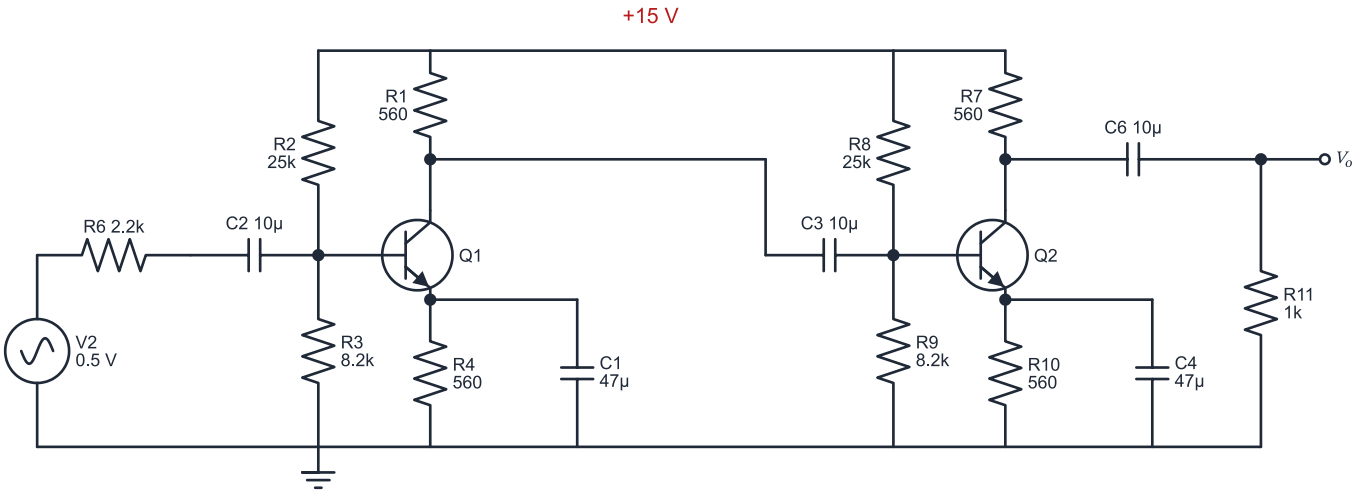


Fig E.2, redrawn: two copies of the Exp 6 stage. C3 couples Q1's collector to Q2's base; the load is R11 = 1 kΩ.

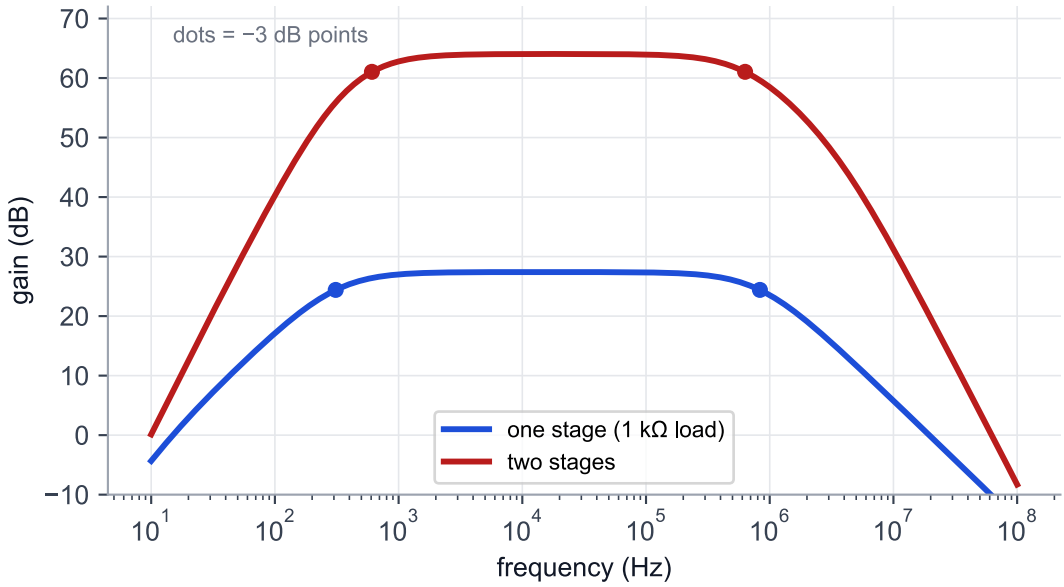
- **Gains multiply:** $A_{total} = A_1 \times A_2$. Decibels therefore **add**: $dB_{total} = dB_1 + dB_2$.
- **Each A must be the gain *in place*.** Stage 1 is loaded by stage 2's input resistance. Stage 2 is driven by stage 1's output resistance (560 Ω) instead of the 2.2 kΩ generator resistance, so it gains more than stage 1 does.
- **Bandwidth shrinks.** At f_L and f_H, *each* stage is already down by some amount, and the losses multiply. So the cascade's −3 dB points move inward: f_L rises and f_H falls.

DO IT (LTSPICE)

1. Draw Fig E.1 (single stage, R5 = 1 kΩ load). Q1 = BC547B, V1 = 15 V, V2 with AC amplitude `0.5` (the manual says 5 V; either works for `.ac`, which is linear).
2. Directive: `.ac dec 10 100 1Meg` (or 500 Hz – 3 MHz as in the procedure). Run, then read the midband dB, f_L and f_H.
3. Copy the stage. Connect C3 from Q1's collector to Q2's base, and put the 1 kΩ load at the end. Run again.
4. Plot both V(b2) (the output of stage 1) and V(out), so you can fill both columns of the table.

TRAP: NO TRANSIENT AT 0.5 V

With a gain of about 1600, a 0.5 V input asks for an 800 V output. Any `.tran` run at that level is pure clipping. Use millivolts for transient, or stick to `.ac`.



LTspice. One stage: 27.4 dB, f_L 310 Hz, f_H 833 kHz. Two stages: 64.0 dB, f_L 609 Hz, f_H 632 kHz.

f (Hz)	100	500	1k	5k	10k	100k	500k	1M	2M	3M
one stage A_v	7.2	19.9	22.4	23.4	23.4	23.3	20.1	15.0	9.0	6.3
two stages A_v	103	990	1379	1584	1591	1575	1252	839	435	264

	Single stage	Two stage
midband gain	23.4 (27.4 dB)	1591 (64.0 dB)
f_L	310 Hz	609 Hz
f_H	833 kHz	632 kHz
bandwidth	≈ 833 kHz	≈ 632 kHz

Gains multiply, in place: inside the cascade at 10 kHz, stage 1 gives 25.8 and stage 2 gives 61.8, so 25.8 × 61.8 = **1594** ≈ 1591 ✓. In dB: 28.2 + 35.8 = 64.0 dB ✓. The stage-alone figure (23.4) is not the in-place gain, so $23.4^2 \neq 1591$.

WRITE IN YOUR RECORD

Result: two-stage gain **64.0 dB** (1591) against **27.4 dB** for one stage; bandwidth falls from ≈ 833 kHz to ≈ **632 kHz** (f_L = 609 Hz, f_H = 632 kHz). Higher gain, narrower bandwidth.

Theory for the record: when amplifier stages are cascaded, the overall voltage gain is the product of the individual (loaded) stage gains, so gains in dB add. In RC coupling, the coupling capacitor passes the AC signal from one stage's collector to the next stage's base while blocking DC, so each stage keeps its own bias. It is cheap and gives a flat midband response. The overall bandwidth is narrower than a single stage's, because each stage's roll-off multiplies: the cascade's lower cut-off rises and its upper cut-off falls.

Part 2 · The design problem (manual p. 3)

MANUAL P. 3 · DESIGN EXPERIMENT

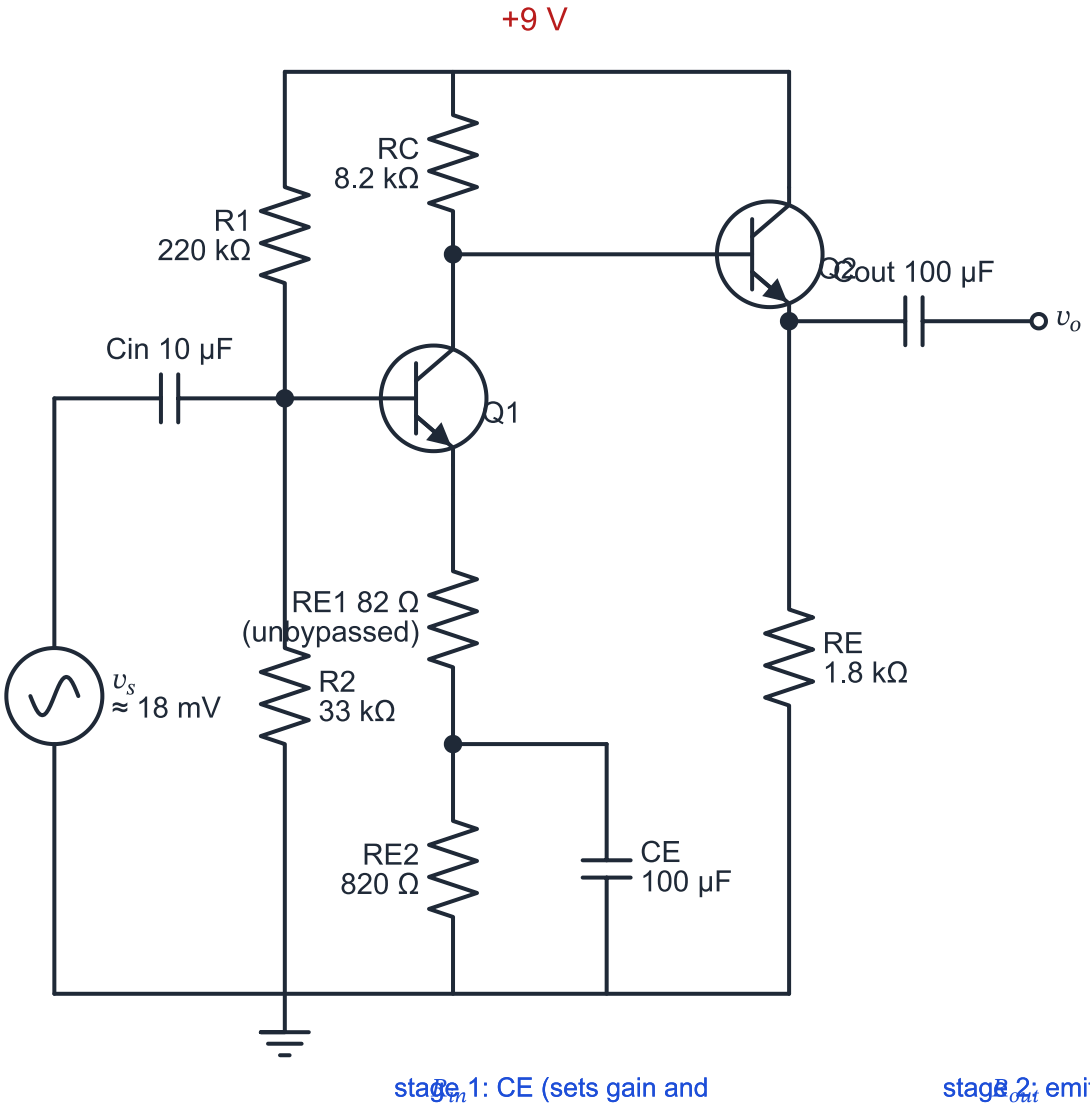
Design a small-signal voltage amplifier with: (1) voltage gain ≥ 50; (2) input resistance ≥ 15 kΩ; (3) output resistance < 100 Ω; (4) maximum unclipped output of 2 V p-p; (5) 9 V supply and low current drain.

GUESS FIRST

Which single stage from Exps 6–8 meets *both* "gain ≥ 50" and "R_{out} < 100 Ω"? If none, what combination does?

The reasoning, one spec at a time:

1. **Gain ≥ 50** calls for a CE stage. But a fully bypassed CE stage has a low R_{in} (≈ βr_e, about 1 kΩ in Exp 6).
2. **R_{in} ≥ 15 kΩ:** leave a small part of R_E **unbypassed** (R_{E1}). The base then sees $\beta(r_e + R_{E1})$, which is much larger, at the cost of some gain (Exp 8, partly). Use large bias resistors.
3. **R_{out} < 100 Ω:** a CE output is ≈ R_C (kilohms), so it fails. Add an **emitter follower** as stage 2 (Exp 7): R_{out} ≈ $r_{e2} + R_C/\beta$, which is tens of ohms.
4. **Low current drain:** run stage 1 at about 0.6 mA and stage 2 at about 2 mA.



Verified design. Stage 1 is a CE with R_{E1} = 82 Ω unbypassed; stage 2 is an emitter follower direct-coupled from Q1's collector. Both BC547B.

Hand design:

- **Stage 1 bias:** V_{B1} = 9 × 33/(220 + 33) = 1.17 V, V_{E1} ≈ 0.5 V, I_{E1} = 0.5/(82 + 820) ≈ 0.55 mA, r_{e1} = 26/0.55 ≈ 47 Ω.
- **Stage 1 gain:** ≈ $R_C/(r_{e1} + R_{E1})$ = 8.2k/(47 + 82) ≈ **64**.
- **R_{in}:** = 220k ∥ 33k ∥ β(r_{e1} + R_{E1}) = 28.7k ∥ (290 × 129 Ω = 37k) ≈ **16 kΩ** ✓.
- **Stage 2:** V_{C1} ≈ 9 − 0.55m × 8.2k ≈ 4.4 V, so V_{E2} ≈ 3.7 V, I_{E2} = 3.7/1.8k ≈ 2 mA, r_{e2} = 13 Ω. R_{out} ≈ 13 + 8.2k/290 ≈ **41 Ω** ✓.
- **Swing:** V_{E2} = 3.7 V sits mid-supply, with room for ±1 V and more ✓.

LTspice check of every spec (`src/sim/design.py`):

Spec	Required	Simulated	
voltage gain (1 kHz, 10 kΩ load)	≥ 50	**59.7**	✓
input resistance	≥ 15 kΩ	**16.2 kΩ**	✓
output resistance	< 100 Ω	**39 Ω**	✓
unclipped output	2 V p-p	**2.95 V p-p** clean (±1.43/−1.52 V at 25 mV in)	✓
supply / current	9 V, low	9 V, **2.68 mA** total	✓
bandwidth (bonus)		14 Hz to 4.4 MHz	

CHECK 1

(a) Two stages have midband gains of 20 dB and 26 dB in place. Give the overall gain as a ratio. (b) Why not reach "gain ≥ 50" with a fully bypassed emitter in the design? (c) In the design, which single resistor would you change to raise the gain, and what does it cost?

Answers

Part 1 question. Gains **multiply** (dB add). Bandwidth **shrinks**: f_L rises and f_H falls.

Part 2 question. No single stage meets both specs: CE has gain but high R_{out}, and CC has low R_{out} but gain < 1. So use **CE followed by CC**.

Check 1. (a) 46 dB = $10^{46/20} \approx$ **200**. (b) The input resistance would drop to about $\beta r_e \approx 13$ kΩ in parallel with the bias network (and lower at higher current), failing the 15 kΩ spec. The unbypassed R_{E1} is what lifts R_{in}. (c) Lower R_{E1} (or raise R_C). Lower R_{E1} raises the gain but lowers R_{in} (back towards the 15 kΩ limit) and adds distortion. Raising R_C raises the gain but lowers V_{C1}, which eats into the output swing.

16 · Revision set: one question per idea, for the night before

Answer from memory on paper, then check the answers page. Any miss points back to its chapter, shown in brackets.

Draw from memory

1. The zener regulator (Exp 2) with values, the clamper (Exp 4), and the CE amplifier (Exp 6) with every part labelled. Then say each part's job in one line.

Diodes

2. A DSO shows 6 divisions peak-to-peak at 0.5 V/div on a sine. Give V_{rms}. *(Exp 0)*
3. Zener: define dynamic resistance, and say why "dynamic". *(Exp 1)*
4. A 6.2 V zener: which breakdown mechanism, and why is its temperature drift small? *(Exp 1)*
5. In the zener regulator, V_{in} goes up by 3 V. Which part takes almost all of it? *(Exp 2)*
6. Write the formulas for % line regulation and % load regulation. *(Exp 2)*
7. A positive clipper uses a 3 V battery. Clipping level? *(Exp 3)*
8. Clipper vs clamper: what happens to V_{pp} in each? *(Exps 3–4)*
9. What condition on RC makes a clamper work at 1 kHz? *(Exp 4)*

BJT

10. Bias of the two junctions in active, saturation and cut-off. *(Exp 5)*
11. Name the four h-parameters, and say from which curve, with what held constant, each is read. *(Exp 5)*
12. In Exp 6, which capacitor mostly sets f_L, and why does the gain fall at high f? *(Exp 6)*
13. Define bandwidth and the −3 dB point. *(Exp 6)*
14. Why is the CE output 180° out of phase? *(Exp 6)*
15. Emitter follower: gain, phase, input and output resistance (high/low). *(Exp 7)*
16. Removing C_E: name the type of feedback, and two things that improve. *(Exp 8)*

FET, oscillator, MOSFET

17. Define g_m, r_d, μ and show $\mu = g_m r_d$. *(Exp 9)*
18. Write Shockley's equation, and define I_{DSS} and V_P. *(Exp 9)*
19. CS amplifier gain formula, and why the gate divider can use megohms. *(Exp 10)*
20. Barkhausen criterion, and the frequency formula of the RC phase-shift oscillator. *(Exp 11)*
21. Why must the amplifier's gain be at least 29 in the phase-shift oscillator? *(Exp 11)*
22. Resistive-load inverter: the MOSFET's state and the output for each input. Define V_M. *(Exp 12)*
23. Two cascaded stages: what happens to the gain and to the bandwidth? *(Extra)*
24. In LTspice, how do you enter 1 MΩ, and which analysis gives a frequency response? *(01)*

Answers

1. See each chapter's redrawn figure and part table.
2. V_{pp} = 3 V, so V_p = 1.5 V and V_{rms} = 1.5/√2 = **1.06 V**.
3. $r_z = \Delta V_Z/\Delta I_Z$ in breakdown: the slope of the curve, not V/I at one point.
4. A mix of tunnelling (Zener) and avalanche. Their temperature coefficients are opposite (− and +), so they nearly cancel.
5. The series resistor R1. The extra current goes into the zener, and V_o barely moves.
6. Line: $\Delta V_o/\Delta V_{in} \times 100$. Load: $(V_{NL} - V_{FL})/V_{FL} \times 100$.
7. 3 + 0.7 = **3.7 V**.
8. A clipper reduces V_{pp}. A clamper keeps V_{pp} and shifts the DC level.
9. RC ≫ T (= 1 ms). Here RC = 10 ms.
10. Active: BE forward, BC reverse. Saturation: both forward. Cut-off: both reverse.
11. $h_{ie} = \Delta V_{BE}/\Delta I_B$ (I_B fixed). $h_{re} = \Delta V_{BE}/\Delta V_{CE}$ (I_B fixed). $h_{fe} = \Delta I_C/\Delta I_B$ (V_{CE} fixed). $h_{oe} = \Delta I_C/\Delta V_{CE}$ (I_B fixed).
12. The emitter bypass C_E. At high f, the transistor's internal junction capacitances (Miller effect) and stray capacitance shunt the signal.
13. BW = $f_H - f_L$, between the frequencies where the gain is 0.707 × max = −3 dB (half power).
14. A rising base voltage raises I_C, which increases the drop across R_C, so the collector voltage falls.
15. Gain ≈ 1 (slightly less), in phase; input resistance high, output resistance low; used as a buffer.
16. Current-series negative feedback. Gain becomes stable (independent of β), bandwidth widens, distortion and noise fall, input resistance rises.
17. $g_m = \Delta I_D/\Delta V_{GS}$ (V_{DS} const), $r_d = \Delta V_{DS}/\Delta I_D$ (V_{GS} const), $\mu = -\Delta V_{DS}/\Delta V_{GS}$ (I_D const) = $(\Delta V_{DS}/\Delta I_D) \times (\Delta I_D/\Delta V_{GS}) = r_d g_m$.
18. $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$. I_{DSS} = drain current at V_{GS} = 0 in saturation. V_P = the V_{GS} at which $I_D \to 0$.
19. $A_v = -g_m(R_D \parallel R_L)$. The gate draws no current, so megohm resistors drop no voltage.
20. $|A\beta| \geq 1$ and loop phase 360°. $f = 1/(2\pi RC\sqrt{6})$.
21. At the oscillation frequency, the three-section RC ladder attenuates the signal to 1/29.
22. Input low: MOSFET off, output = V_{DD}. Input high: MOSFET on, output ≈ 0. V_M = the input voltage at which $V_{out} = V_{in}$.
23. Gains multiply (dB add). Bandwidth shrinks.
24. `1Meg` (`1M` is milli). `.ac`.